Refractive Gravity: An Ontological Picture and the Foundations of Mathematical Formalism

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Abstract

Refractive Gravity (RefG) begins from a simple operational fact: the Universe cannot be measured from outside, because clocks, rulers, light, and observers are all made of the same physical world. Building on this fact, RefG develops a unified ontology in which one universal pre-spatial foundation, through resonant self-distinction, gives rise to operational space, time, matter, and gravitational geometry. Stable massive matter is represented by locally self-confined nonlinear wave structures—oscillons—whose dynamics establish pressure-deficit profiles in the manifested foundation-medium. The local ruler scale, externally read mass, and clock cadence follow a common factor p, while coordinate light propagation follows p^2. In the post-Genesis relational-coframe continuum, the selected leading low-energy class yields the TEGR combination; the exact torsion–curvature identity then converts the action to Einstein–Hilbert form up to a boundary term. Under the complete stated 1PN conditions, the standard PPN vector follows. The strict-vacuum quadratic sector of the oscillon amplitude–phase action takes exactly the form of the free complex Klein–Gordon action on that same coframe; on the fixed Minkowski branch, standard canonical quantization yields positive-energy particle and antiparticle sectors of bosonic Fock space, with mass m and spin zero for the free excitations about the strict vacuum. Cosmologically, growth of the foundation's inter-node scale lowers its background pressure and the material scale, while an internal observer reads this single causal history through $A = a/p$. The monograph presents this physical picture as a continuous intuitive argument; its formal calculations are archived on Zenodo, and the exact status of every sector is collected in the final section.

Contents

Preface

We cannot measure the Universe from outside it. The clock, the ruler, light, and the observer are all constituents of the same physical world. Refractive Gravity (RefG) begins with this fact and asks: what does a universe see when it measures itself with its own matter? From that question unfolds a unified ontological picture of one foundation and of the geometry, matter, and time that arise from it. In the specified low-energy limit, the same relational coframe carries Einstein–Hilbert geometry, while the strict-vacuum quadratic sector of the oscillon action is read on it as a free complex scalar quantum field. The text is structured as an intuitive journey; the corresponding formalism and calculations are available in the public scientific archive on Zenodo [Lotishvili, 2026a,b].

The subjects developed here stand at different stages of the research program. Some have been obtained and tested formally; others are physically specified constructions or mathematical directions opened by the same picture. Their exact status is collected in the final section, "Current Status of the Research Program." This arrangement preserves the continuity of the argument while presenting the technical boundaries clearly in one place.

Introduction

Modern fundamental physics rests on a monumental yet internally divided architecture. Einstein's General Relativity describes gravity as the dynamical geometry of spacetime. Quantum field theory, by contrast, usually represents matter and gauge forces as operator excitations on a prespecified background.

Both pillars have been confirmed with extraordinary experimental precision. Their ontological starting points nevertheless speak different languages: gravity itself creates the physical arena, whereas quantum theory formulates dynamics with respect to an already defined temporal parameter.

Against this profound conceptual divide, contemporary physics confronts a cluster of unresolved foundational crises:

1. **The Singularity Problem:** Classical descriptions of black holes and the cosmological origin lead to geodesic incompleteness and, in important cases, curvature divergence; this is where the physical reach of continuous classical geometry ends.

2. **Astrophysical and Cosmological Anomalies:** Within the standard cosmological model, galactic rotation curves, the Radial Acceleration Relation (RAR), and the Baryonic Tully–Fisher Relation (BTFR) are described through dark-matter halos together with baryonic galaxy-formation physics, while accelerated cosmic expansion is attributed to a vacuum-energy cosmological-constant term. A naive continuation of field theory to a Planck-scale cutoff may exceed the observed vacuum density by roughly 120 orders of magnitude.

3. **Structural and Hierarchical Puzzles:** Quantum-measurement randomness, the nature of the arrow of time, the pronounced hierarchy of charged-lepton and quark masses, the unexplained values of dimensionless constants—including $\alpha \approx 1/137$—and the extreme weakness of gravity relative to the electroweak scale ($M_{\mathrm{Pl}}/v_{\mathrm{EW}} \sim 10^{17}$) remain disconnected inputs or unresolved structures in the standard paradigm.

The history of physics shows that apparently separate paradoxes often bear the trace of a deeper common foundation. Einstein pursued a geometric unification of electromagnetism and matter throughout his later life. John Archibald Wheeler explored the boundary between the geometry and information of the Universe and gave that search two enduring formulations—*Mass without Mass* and *It from Bit* [Wheeler, 1955, 1990].

Andrei Sakharov read gravity as an elastic effect induced by quantum vacuum fluctuations [Sakharov, 1967]. Ted Jacobson derived Einstein's equations from the thermodynamic equation of state of local horizons [Jacobson, 1995]. Erik Verlinde described attraction as a holographic entropic force [Verlinde, 2011], while Roger Penrose connected the causal structure of spacetime to gravitational collapse through the singularity theorem [Penrose, 1965].

While these frameworks employ distinct mathematical vocabularies, they share a singular, foundational question: **Is it possible that the entire tapestry of physical phenomena—space, time, gravity, and the particle spectrum—is the dynamic manifestation of a single primary ontological substrate?**

This monograph answers this question in the affirmative, presenting the unified conceptual and ontological framework of **Refractive Gravity** (RefG).

Operational Foundations and Three Fundamental Principles

The standard language of physics typically begins with an established measuring apparatus: smooth coordinates, a continuous time axis, and operator fields are treated as axiomatic primitives. RefG grounds its inquiry in the strict operationalism of Percy Bridgman [Bridgman, 1927]: *Before speaking of space, time, or mass, what physical process generates the local, distinguishable differences upon which rulers and clocks function?*

In 1905, Einstein dismantled the concept of absolute time by operationally defining simultaneity via clocks and light signals [Einstein, 1905]. RefG pushes this operational critique to a more primordial level: **What physical mechanism constructs the stable clock, the material particle, and the propagating wave signal upon which all such measurements fundamentally depend?**

In response to this question, RefG formulates three fundamental postulates:

1. **Monistic Ontological Foundation:** Nature possesses one universal, pre-spatial, and pre-clock foundation. Its resonant self-distinction generates a manifested medium whose phase, pressure, shear, rotational, and topological modes appear operationally as elementary fields, forces, and particles.

2. **The Oscillon Nature of Matter:** A stable massive material unit is an *oscillon*—a locally self-confined, self-sustaining nonlinear wave configuration of the manifested medium. Massless carrier modes are the freely propagating radiation waves of the same medium. The oscillon's intensive internal oscillation establishes a persistent static pressure-deficit profile in the surrounding medium.

3. **Gravity as Effective Refraction:** Oscillon matter establishes an inhomogeneous pressure-deficit landscape throughout the medium. This deficit alters the effective geometry, causing the wavepaths of light and matter to bend refractively. Gravitational attraction is thus revealed as the physical mechanism of spacetime curvature. In the relational-coframe description, the pressure deficit is the volumetric scalar readout of the full local geometry. Symmetric shape and shear determine the tensor structure of gravity, while frame orientation is an inertial gauge freedom.

Lorentz Invariance and Historical Precedents

The Michelson–Morley experiment of 1887 detected no "aether wind" from the Earth's orbital motion, placing stringent constraints on 19th-century models of a rigid, matter-independent "luminiferous aether" [Michelson and Morley, 1887]. The RefG foundation-medium belongs to an entirely different ontological category: here, measuring rods, clocks, the observer, and propagating light are themselves collective wave excitations woven from this very medium.

When observer and apparatus are woven from the same medium, uniform translational motion is a symmetric phase translation of the wave form. On the Lorentz-covariant branch this motion is lossless and leaves no aether wind. Lord Kelvin's intuition of vortex atoms thereby acquires the language of modern nonlinear wave dynamics [Thomson, 1867]: **matter is an organized form of motion of the medium itself.**

Eight Pillars of the Framework

This text establishes the ontological architecture of RefG and connects the computational formalism archived on Zenodo to its physical meaning. Medium-based unifications have notable precedents, including Winterberg's Planck-aether model [Winterberg, 2002]; the distinctive starting point of RefG is the birth of measurability itself and the population-level resonant stabilization of matter.

The unified RefG picture rests on eight pillars:

1. **The Birth of Time and a Common Oscillon Cadence:** In the Genesis scenario of RefG, time begins with a centerless transition encompassing the whole foundation. The first distinguishable relations create event order; within the resulting network, oscillon seeds and later stable matter take form. A local pressure change transposes the particle spectrum through the common cadence Ω_t, while the oscillon's internal structure is retained in the dimensionless labels s_i and their ratios.

2. **Two-Channel Mass Accounting:** The oscillon's internal topological state describes its conserved energetic store; externally read gravitational mass is the integral response of the pressure deficit established in the medium. Separating these two channels prevents the same energy from being counted twice.

3. **Biconformal p/p^2 Scaling and the Complete 1PN Response:** The external footprint of a local ruler, externally read mass, and clock cadence follow p, whereas coordinate light speed follows p^2. In the static spherical weak field this geometry gives $\gamma = 1$. With a canonical gradient term, linear coupling to the active source density entering Gauss's law, and no additional unsuppressed term at the same order, the logarithmic response $u = -\ln p$ also gives $\beta = 1$. In the corresponding low-energy class of the post-Genesis relational-coframe action, the TEGR coefficients are obtained, and the exact torsion–curvature identity converts the action to Einstein–Hilbert form. When the complete 1PN conditions are satisfied, the standard PPN vector follows.

4. **One Coframe for Gravity and Free Quantum Matter:** The vacuum quadratic sector of the nonlinear oscillon amplitude–phase action takes exactly the form of the free complex Klein–Gordon action on the same relational coframe. On the fixed Minkowski branch, standard canonical quantization of this quadratic sector yields positive-energy particle and antiparticle sectors of bosonic Fock space, a conserved global phase charge, microcausality, mass m, and spin zero for the free excitations about that vacuum. Einstein–Hilbert geometry and standard free quantum matter are thereby placed on one operational geometry.

5. **The Koide C_3 Phase Geometry:** A three-phase map in charged-lepton square-root mass space reproduces Koide's $K = 2/3$ identity exactly when $A_K = \sqrt{2}$. RefG reads this geometry as a candidate imprint of a ninth-order return cycle in which $h = 2$ fixes the phase angle $\theta = 2/9$.

6. **The Nodal Origin of the Fine-Structure Constant (α):** The fine-structure constant $\alpha \approx 1/137$ is read as the dimensionless ratio between the electric-flux and transverse-photon responses of one nodal network. Both channels transform together, leaving their ratio invariant.

7. **Genesis, Foundation Expansion, and Its Internal Readout:** The first distinguishable relations begin time; resonant coupling connects the network; stable matter, radiation, and long-term relaxation take form in subsequent phases. In the post-Genesis realization, growth of the foundation's inter-node scale lowers P_F, and the pressure decline reduces the material scale p. A neutral collective phase charge is distributed through the expanding three-dimensional volume: its conservation gives $\eta_F a^3 = 1$, while homogeneous constitutive relations yield $P_F/P_{F,0} = a^{-3}$, $p = a^{-3/2}$, and $A = a/p = a^{5/2}$. Accordingly, the internally measured A uniquely reconstructs the normalized relative scales: $a = A^{2/5}$, $p = A^{-3/5}$, and $\eta_F = A^{-6/5}$. The variables p_C and P_F describe distinct physical quantities; their constitutive relation belongs to the microphysics of the foundation. On the ideal homogeneous background, A unifies redshift, signal time dilation, and geometric distances in one response. When the cold phase source couples to the plasma

only through the common geometry and the local dimensionless atomic and QED ratios retain their standard values, the CMB initial-value problem on this same A geometry takes exactly the standard Einstein–Boltzmann, recombination, and line-of-sight form.

8. **Two Readouts of a Single Spectral Operator—A Central Hypothesis:** The short-wavelength eigenfunctions of a unified volumetric Helmholtz/Laplace-type resonator are read as microscopic particle modes, whereas its long-wavelength modes are read as the macroscopic structure of the cosmic web and galaxy clusters.

Historical Lineage and Priority Boundaries

Every component of RefG is anchored in the established heritage of theoretical physics:

- The p/p^2 scaling law is related to the Polarizable Vacuum (PV) approaches of Robert Dicke and Harold Puthoff [Dicke, 1957, Puthoff, 2002]; RefG ties it to the ontology of a nodal network and two-channel mass accounting.
- The three-phase geometric reading of Koide's formula extends the work of Carl Brannen and Robert Foot [Brannen, 2010, Foot, 1994], while the holonomic origin of 2/9 resonates with the closed family cycles studied by Konstantin Shulga [Shulga, 2026]. RefG combines this structure with the θ-blindness argument and the C_3 symmetry of a three-dimensional volumetric resonator.
- The historical motivation for background-pressure relaxation and a possible microscopic origin of Λ comes from the q-theory of Klinkhamer and Volovik [Klinkhamer and Volovik, 2008]; RefG writes the background dynamics of its selected branch in Einstein–Hilbert geometric language. The motivation for three spatial dimensions is connected with studies of topological knotting and stability in nodal structures [Whitrow, 1955, Berera et al., 2017].

The defining contribution of this monograph is the synthesis of these disparate threads into a single, logically connected research program: tracing the path from the operational birth of measurability and oscillon mass to galactic vortex dynamics and cosmological horizons.

The text invites the reader to follow this chain and see how a unified physical intuition unfolds into a broad panorama of contemporary fundamental physics.

1 The Ontological Foundation

1.1 A Single Foundation and Its Manifested Medium: Substance, Void, and the Dynamics of Existence

Modern fundamental physics describes the Universe as a system of distinct, interacting quantum fields. The Standard Model organizes these fields through shared gauge symmetries, internal degrees of freedom, and interaction rules. Refractive Gravity (RefG) sees one fundamental principle beneath this diversity: **Nature possesses one universal, pre-spatial ontological foundation.** Its resonant self-distinction generates a measurable, multichannel medium. Distinct fundamental fields, gauge forces, and elementary particles are read as the topological, phase, pressure, shear, and rotational modes of this manifested medium. Throughout this monograph, the term "foundation-medium" denotes this self-distinguished, physically manifested state of the primary foundation.

This monistic vision possesses deep historical roots in theoretical physics. As early as 1870, William Kingdon Clifford advanced the prescient hypothesis that matter consists of small, propagating ripples of spatial curvature [Clifford, 1876]. RefG endows this geometric intuition with a concrete physical mechanism: the self-distinguished medium supports not only geometric curvature responses, but also pressure, shear, hydrodynamic, and resonant channels.

A clear theoretical precedent for the emergence of distinct particle and gauge sectors from one carrier is the string-net condensation model of Michael Levin and Xiao-Gang Wen [Levin and Wen, 2005]. In that framework, collective many-body entanglement produces structural analogues of gauge bosons and fermionic excitations. Levin–Wen begins with a prestructured quantum lattice; RefG associates the same unifying potential with the self-distinction of one universal pre-spatial foundation.

Conceptualizing the Universe as a unified continuum requires two asymptotic dynamical bounds:

1. **The Absolute Maximum of Substance:** A saturation limit characterized by maximal pressure, mass–energy potential, and operational nodal density.

2. **The Zero-Distinguishability Bound:** A limit in which interaction and physical measurability vanish entirely.

This "void" is the logical limit at which physical distinguishability vanishes. Objective physical reality is the bounded, nonlinear oscillatory dynamics between these two poles, within which stable local units of the foundation—"nodes of existence"—form.

The reference background $\varphi = 0$ is the quiescent yet physically active state situated between these two extrema. At the limit of zero distinguishability, nodal interactions and measurable traces completely vanish; $\varphi = 0$ designates the unperturbed baseline state of the manifested medium.

The ontological unity of the manifested medium is accompanied by a rich multi-channel capacity. The medium can compress (volumetric pressure), undergo topological twisting, establish vortical circulation, and support transverse wave excitations. A hydrodynamic analogy provides useful intuition: a single fluid concurrently supports isotropic hydrostatic pressure, vortex circulation, shear stress, and surface waves. Similarly, the RefG manifested medium integrates phase, pressure, shear, and topological responses, providing the dynamic substrate upon which electric charge, spin, and the particle spectrum are constructed.

The foundation-medium's unperturbed state, formally denoted by $\varphi = 0$, is its background vacuum regime. The foundation is ubiquitous; only the degree of excitation, the operational density of distinguishable nodes, and the strength of their coupling vary locally. Growth of the operationally inaccessible intervals between nodes rarefies the foundation and lowers its pressure state.

Here pressure is the effective pressure–energy state variable of RefG. The unperturbed background is $\varphi = 0$; $\varphi > 0$ is the elevated pressure–energy branch (compression), and $\varphi < 0$ is the deficit branch (rarefaction). The gravitational readout is built on the gradient profile of the deficit branch. Classical Bernoulli pressure is its immediate physical analogue: an external observer reads a local change in φ as a change in clock cadence, coordinate signal propagation, operational size, and effective gravitational mass.

1.2 From Single Existence to Multiplicity: The Emergence of Measurability and Operational Space

In the Genesis postulate of RefG, the foundation's initial state is wholly undifferentiated: change, event order, duration, and geometry have not yet arisen. The birth of the Universe is the first physical event and the origin of temporal reckoning.

The most coherent origin scenario in RefG is a centerless transition encompassing the whole foundation. It is one shared initial condition without a distinguished spatial center. Space, physical distance, propagation speed, and an "outside" acquire meaning only after the relations created by this transition have formed.

At this boundary the foundation first produces minimally distinguishable relations. Event order and both measures of time begin here, at $t = 0$ and $\tau = 0$. This initial layer of distinguishable connections is the minimal network of existence; material particles and stable oscillons form during its subsequent development.

Nonlinear modes localize on the first relations as oscillon seeds. Their resonant fields spread across the relational layer and overlap. Once the physical threshold of distinguishability is crossed, this overlap binds operationally separate regions into a single network. The causal sequence therefore begins with the first relation and continues through localization and network coupling. Local locking and global coupling may be partly contemporaneous; their precise ordering is set by resonant dynamics and the physical coupling threshold.

The Genesis picture presents the Universe as an interconnected network with neither a boundary nor a distinguished center. A finite periodic graph provides a simple mathematical image of this arrangement: the network remains finite and connected without singling out any node as a unique center. The structure of the foundation's relations determines the dimension and topology of the physical Universe.

Energy in the initial coherent state is distributed among oscillon seeds, their resonant fields, freely propagating waves, a thermal sector, and the collective deformation of the foundation. A continuity law gathers the sectoral distribution of energy and the fluxes between sectors into one balance: a loss in one sector appears as a gain in another. Phase energies and the equation of state determine the physical direction of this redistribution. The *melting* of ice is the intuitive image of the transformation.

The first collective change initiates the foundation's self-distinction. Operational space becomes measurable when a stable relation between nodes reaches the minimum distinguishability scale ℓ_*. The interval between two distinguishable nodes is the physical embryo of space. Delay in relational signal transfer creates measurable duration, while the local energetic strain of self-distinction is the ontological embryo of mass. The integral response of the deficit profile produced by this strain in the surrounding foundation is later read as gravitational mass.

Physical meaning invariably requires comparison. At the ontological origin, three indivisible elements emerge:

- **Node:** A locally distinguishable, discrete state of the foundation;
- **Trace:** The energetic and phase alteration imparted by a node to the surrounding medium;
- **Relation:** The operational capacity to compare one local state with another.

From this foundational triad unfolds the entire operational network that macroscopically manifests as space, time, and matter.

A completely undistinguished foundation possesses a single, indivisible identity. Its resonant self-distinction generates the first persistent difference, marking the birth of quantity and counting. Distinguishable manifestations are registered as separate physical units without the foundation fragmenting into disconnected parts—multiplicity consists of the stable, mutually distinct states of one fundamental identity.

The primary unit of measurement is a preserved relation. In an undistinguished state, no second entity exists for comparison. Self-distinction yields two distinct states of the same foundation and establishes the primary link between them; here begin counting, spatial extension, and the relative ordering of events.

In information-theoretic terms, the first preserved distinction is analogous to a bit: a distinguishable trace is registered as "1", and its absence within a given relational frame as "0". Complex arrangements of these binary states generate rich macroscopic structures, just as the mutually distinct states of the foundation generate the diversity of the observable cosmos.

The concept of one object existing at different locations at different times illustrates this logic. If the temporal labels along its worldline are removed, a single identity leaves traces at many locations. RefG carries this intuition to the pre-spatial level: persistent self-distinctions of the foundation generate the locations that can then be counted and compared. The primary foundation is considered outside time; the age of the measurable Universe belongs to the event order that arises from it.

This perspective is crucial for understanding the nature of the observer. Observers and their measurement instruments are themselves complex, composite wavepackets built from this nodal network. This deepens Percy Bridgman's operationalism, which asserts that physical concepts are defined solely by the operations required to measure them [Bridgman, 1927]. RefG provides this operational philosophy with an ontological substrate, investigating the physical conditions that make measurement operations possible in the first place.

The measurable Universe is read as a continuous process of self-distinction in the foundation-medium. Operational space arises along the trace of this distinguishability. Beyond that trace the universal foundation may persist, but it manifests as a physical object only where it leaves an operational, measurable result.

The absolute separation between nodes cannot be measured directly, because no observer possesses a measuring rod external to the medium. What is measurable is strictly the operational trace of the inter-node interval. Operational space is born when the relation between at least two nodes becomes physically distinguishable. Below this threshold, variation remains an unmanifested internal state; above it, it registers as shifts in clock rates, operational sizes, mass readouts, and signal propagation.

RefG introduces a dynamic lower bound of minimal operational distinguishability, denoted by ℓ_*:

- At scales smaller than ℓ_*, no independent measurable relation can form (information is indistinguishable);
- At the scale ℓ_*, the first stable, physically distinguishable record emerges;
- At scales larger than ℓ_*, inter-node intervals register as changes in pressure, clock rates, and signal propagation.

The physical rationale for this bound is the finite resolution required by a stable record. The Planck length $\ell_{\mathrm{Pl}} = \sqrt{\hbar G/c^3}$ is a dimensional scale constructed from fundamental constants, whereas ℓ_* is the dynamical threshold for stable record formation. Their relation is fixed by the microdynamics of the foundation.

The ontological origin of the Universe is a completely undifferentiated primary foundation-state. A geometric point is already a concept of formed space; two distinguishable points and the extension between them arise together with the operational inter-node interval. An asymmetry formed in the foundation-medium leaves a stable trace, the relation becomes readable, and multiplicity and geometry emerge in the same event.

At this threshold the primary physical meaning of gravity becomes clear: when the pressure state of the foundation-medium has a spatial gradient, clock cadence, operational size, and coordinate signal propagation change coherently, and this change is read as gravitational geometry. A homogeneous change produces common background self-calibration; acceleration and refractive bending are produced by spatial inhomogeneity.

1.3 Nodal Transmission: Waves, Time, and Self-Calibration

The foundation's first collective change creates a minimal relational layer. Each node is a functionally distinguishable state that preserves the trace of a physical change and can relay it to adjacent nodes. Subsequent transmission, together with resonant coupling produced by overlapping resonant fields of oscillon seeds once the distinguishability threshold is crossed, joins this layer into an operational network. Spatial proximity and operational distance emerge from the stable organization of these physical couplings.

A wave in this network is the sequential transmission of state and phase. Light is a freely propagating transverse phase trace, whereas an oscillon is a locally self-confined, self-sustaining configuration of the same transmission mechanism. When matter moves, this localized wavepacket is transferred from node to node, reconstructing its phase integrity at each successive location. Light and matter are thus the propagating and localized modes of a single medium.

The foundation's first complete self-distinction creates the first ordering of events and begins time. Every subsequent transmission within the formed network extends that order. Recurrent phase relations establish a process cadence, while comparisons among stable oscillon cycles make clock measurement possible. The local rate of time is the network's collective rhythm: the rate at which the medium relays state and an oscillon completes its internal cycle.

Thermal insulation in double-pane windows provides a helpful analogy. Rarefying the intermediate gas reduces molecular collisions and slows the rate of heat conduction. RefG extends this intuition to the nodal network, linking the transmission rate directly to nodal density and coupling strength.

The observer's body, clocks, rulers, light, and matter are all modes of this same medium. In a deficit region, nodal density, transmission rates, oscillon cycles, and operational sizes undergo a coordinated shift. Local measuring devices reconfigure synchronously with their environment; consequently, an observer using local standards measures normal clock rates, standard dimensions, and the invariant local speed of light (c). Comparing processes that have traversed different environments, however, reveals objective differences in accumulated time, operational size, and mass.

Along the selected Lorentz-covariant low-energy branch, this transmission is isotropic, non-dissipative, and strictly respects Lorentz symmetry: the local speed of light remains constant, and uniform motion leaves no frictional wake.

1.4 Local Collective Cadence and Resonant Locking

The birth of time and the formation of a local collective cadence are distinct stages. In the Genesis picture of RefG, time begins with the first foundation-wide self-distinction. A local common cadence forms later, when the intrinsic frequencies of stable material oscillons settle into fixed, persistent ratios. In that collective state, the temporal rate is governed by the internal coordination and nonlinear mutual locking of the local network, in a manner analogous to Kuramoto-type synchronization.

Resonant tone and physical clock cadence arise together with the network of relations. The formed medium behaves like an acoustic instrument—a vibrating volumetric resonator whose connection operator and boundary conditions determine the spectrum of admissible modes. The three-dimensionality of this resonator belongs to the physical map of RefG.

The physics of a bell offers a vivid illustration: its sound arises from the normal modes of one elastic body, arranged in a collective pattern of harmonic and anharmonic relations. Likewise, elementary-particle frequencies are bound by mutual balance and coordination within the common dynamics of the foundation. The universal foundation determines the arena of admissible modes, while particle modes balance one another within that arena to form a stable, resonantly locked configuration.

The pressure state changes the collective network cadence measured per unit foundation coordinate time. Denote that cadence by $\Omega_t(x)$; the subscript t emphasizes that the frequency is counted per unit of foundation coordinate time. The common pressure response is

$$\frac{\Omega_t(x)}{\Omega_{t,0}} = p(x), \qquad d\tau = p\,dt. \tag{1}$$

Here τ is process time read by material clocks. A falling p means that fewer material cycles occur within the same coordinate-time interval dt. A local observer uses the co-adjusted τ standard and therefore reads a normal local cadence; the difference appears only when environments or epochs are compared.

The oscillon's internal geometry selects its stable modes through dimensionless structural labels s_i. The common-cadence particle picture combines these modes as

$$\nu_i^{(t)}(x) = s_i\,\Omega_t(x), \qquad \nu_i^{(\tau)} = \frac{\nu_i^{(t)}}{p}. \tag{2}$$

The s_i belong to the oscillon's internal spectral structure, whereas Ω_t is the environmental coordinate-time cadence. A common pressure change transposes the entire spectrum by one factor; process time compensates for that shared change in the local clock readout. The labels s_i and their invariant ratios preserve the distinctions among particle modes.

In a low-pressure region, the coordinate-time cadence Ω_t decreases. Inter-node transfer is more frequent in a dense, high-pressure foundation and less frequent in a rarefied one. Retarded inter-node signal transfer, reduction of the foundation's pressure–energy state, and slowing of Ω_t are three readouts of one network state. Each enters the quantitative causal chain exactly once.

Local clocks and atomic processes co-adjust with their environment, so an observer cannot detect the shift using local standards alone. Common transposition requires dimensionless ratios to remain invariant; the observed stability of the fine-structure constant is one of the sharpest tests of this principle. In this picture, the physical source of particle mass ratios lies in the internal s_i spectrum of oscillons.

The historical precursor to this relational approach is Ernst Mach's principle [Mach, 1883]. RefG gives that principle a local, dynamical form: the common cadence of time is governed directly by the collective state of the local resonant network, while the influence of distant masses must be encoded in that local state. Relationality and local causality are therefore properties of the same local network.

In the RefG framework, the fine-structure constant $\alpha \approx 1/137$ is read as an invariant dimensionless ratio between the electric-flux and transverse-photon responses of the foundation's nodal network. In the common-transposition regime it remains unchanged under local shifts in pressure.

Now consider a wave signal propagating from one pressure regime of the foundation to another—for example, from a gravitational well to a distant observer. The photon follows the operational geometry $A = a/p$, while the receiver compares its phase cadence with a local atomic standard. On the ideal transparent branch this comparison gives

$$\frac{\omega_o}{\omega_e} = \frac{A_e}{A_o}, \qquad 1 + z = \frac{A_o}{A_e} = \frac{a_o/p_o}{a_e/p_e}. \tag{3}$$

The common pressure response of the signal and the atomic standard is already contained in the A geometry; the resulting phase ratio is translated into the local Ω_t cadence at each endpoint.

Ultimately, the rate of time in RefG is the ontological state of the local resonant network. When the medium's pressure changes, the network and every compatible process change by a common factor while their dimensionless ratios remain invariant. Every real measurement occurs within one and the same physical network; the processes of that network therefore provide the physical content of time.

This principle establishes a rigorous empirical falsification criterion: **If any dimensionless ratio changes under a local pressure shift—for example, if a reliably established nonzero $\Delta\alpha/\alpha$ is found—the postulate of a population-locked collective cadence is falsified.**

1.5 Oscillons, Pressure Deficits, and the Origin of Mass

RefG reads the electron, proton, and quark as **oscillon modes**. An oscillon is a locally self-confined, self-sustaining nonlinear wave structure in the universal foundation-medium—a highly organized, coherent oscillatory state of the medium itself.

The historical precursor of this concept is Lord Kelvin's hypothesis of "vortex atoms" [Thomson, 1867]. Its enduring intuition is that a particle is a self-confined dynamical topological form of the medium. The oscillon concept develops this idea in the language of modern nonlinear field theory and topology.

When an oscillon oscillates intensely, rotates, or confines energy within its localized geometry, the static pressure of the surrounding foundation-medium decreases. Daniel Bernoulli's hydrodynamic balance provides a clear physical analogue [Bernoulli, 1738]: **Under the appropriate conditions for stationary flow, an increase in the dynamical contribution accompanies a decrease in static pressure.** On the weak, static gravitational branch of RefG, this analogue corresponds to the schematic pressure–energy relation:

$$\Delta P \propto -\rho_{\mathrm{dyn}}. \tag{4}$$

Through its internal oscillation, an oscillon establishes a stable gradient profile of static pressure deficit in the surrounding foundation. In the RefG map, this is the **external profile of gravitational mass**: its volumetric integral gives the system's total gravitational charge. The more intense the oscillon's internal dynamics, the deeper the surrounding deficit and the stronger the refractive gradient. The oscillon's internal energetic store is accounted for in a separate, independent channel.

The core of an oscillon is localized, but its oscillatory trace extends throughout the connected medium. The component phase-locked to the source establishes a static gravitational profile, requiring zero net outward energy flux. When the source undergoes rapid dynamic reconfiguration, the coherent oscillatory component released from this channel propagates outward as a gravitational wave, carrying energy and momentum.

The local depth of the deficit and the asymptotically measured total mass of a system are distinct physical quantities. The former characterizes central gradient and compactness; the latter integrates the active core, foundation-medium dressing, binding energy, and radiated flux into a single asymptotic balance. Following a merger, the deficit may deepen while the total externally read mass of the remnant decreases in accordance with the radiated energy.

This perspective has a well-known historical parallel in John Archibald Wheeler's *Mass without Mass*: a "geon" is a massive formation confined by the energy of its own gravitational and electromagnetic fields [Wheeler, 1955]. Classical Wheeler geons are unstable and gradually radiate energy; RefG associates oscillon stability with conservation of topological charge, resonant locking, and the topological invariant $Lk = Wr + Tw$.

The term "effective refractive mass" (m_{eff}) therefore denotes the externally read gravitational mass of an individual oscillon in a specified background medium. Under slow background variation, the oscillon's topological structure, internal energy, and phase ratios preserve their identity, while its operational scale and externally read mass follow the new state of the medium. Rapid reconfiguration is a distinct dynamical process that redistributes energy among other compatible channels. The total asymptotic mass of a system belongs to a separate integral readout and does not follow automatically from the m_{eff} of a single oscillon. With this distinction, the origin of mass is read as the formation of the medium's refractive response.

At this foundational level, **gravitational attraction is the refractive bending of one oscillon's wavepacket within the pressure deficit generated by another.** A relaxed string provides an intuitive picture of a gravitational well: locally diminished pressure–energy weakens the effective wave-propagation stiffness of the foundation-medium. The deficit profile defines the effective geometry; in the ray limit, the phase trajectories of light and matter wavepackets follow the corresponding null and timelike geodesics of that metric.

This defines the central dynamical reciprocity proposed by RefG: **An oscillon creates a pressure deficit through its internal oscillation, while this deficit determines its effective mass, operational scale, and trajectory through the medium.** Matter dictates the pressure state of the medium, and the altered medium dictates the refractive geometry of matter.

2 Gravity as Refraction

2.1 Three Simultaneous Operational Shifts

The inter-node interval of the foundation-medium is read from within the Universe through its physical trace. Around an oscillon that trace is a local reduction of the pressure state: the density of distinguishable nodes falls and informational or wave transfer between them slows. A macroscopic observer records this one state through three mutually consistent changes:

1. **Reduction of the Coordinate Speed of Light:** In a pressure-deficit zone, nodes are rarefied and the rate of inter-node transmission drops. Consequently, the propagation of light and all other massless wave signals is retarded in coordinate description. The locally measured speed of light remains strictly invariant (c) for a local observer; what changes is the effective propagation rate evaluated in the coordinate language of a distant observer.

2. **Contraction of Oscillon Operational Size:** While the oscillon's intrinsic topological form—its nodal winding configuration—remains invariant, fewer interacting nodes occupy its operational volume in the rarefied foundation-medium. In a distant observer's coordinate description, this is registered directly as a contracted operational size and a modified effective-mass readout.

3. **Retardation of the Local Resonant Clock Cadence:** Real physical clocks are composed of oscillon systems (atoms, molecules) whose internal tick rate depends directly on local nodal coupling. In a rarefied environment, this coupling weakens, decelerating the clock rate in coordinate time relative to an asymptotic observer. Locally, the clock continues to measure its own cadence normally and invariantly.

The rarefied gap of a double-walled thermos provides a physical analogy: reducing the number of carrier molecules lowers the collision frequency and slows thermal transfer. In the RefG map, this corresponds to the relation between nodal coupling strength and coordinate clock cadence.

These three shifts are coordinated operational readouts of one underlying pressure deficit—not independent physical effects. Their quantitative relation establishes the foundation of biconformal geometry.

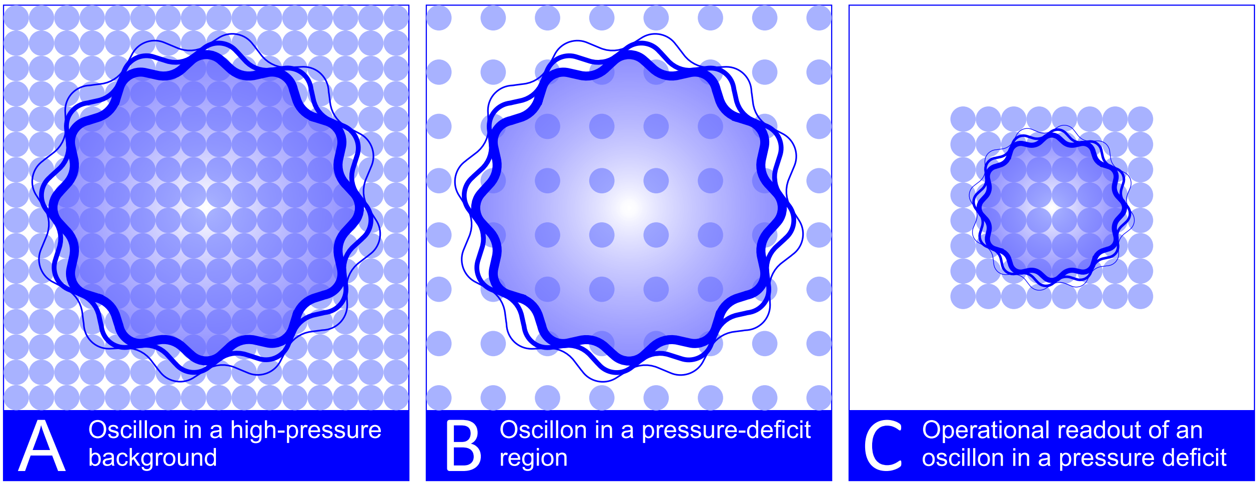

Figure 1: **Operational readout of an oscillon within a pressure-deficit zone. (A)** In an unperturbed foundation-medium whose pressure is higher than in the deficit zone, an oscillon occupies a comparatively dense nodal arrangement; inter-node coupling is stronger and local processes proceed at a higher coordinate cadence. **(B)** Within a pressure-deficit zone, foundation nodes are rarefied; operational inter-node separation increases, transmission cadence falls, and the local passage of time slows. **(C)** In external readout, an oscillon within a pressure deficit is registered with a slower clock cadence, smaller operational size, and changed effective mass; the increased internal interval does not enter as an independently measurable quantity.

The inter-node interval is an internal relation of the foundation-medium. In the macroscopic description, its growth is read as a reduction in the pressure state; this is the operational trace of foundation rarefaction.

2.2 Quadratic Coupling and the Factor of Two

The quadratic relation arises from two coordinate readouts of the same local standard. The external footprint of a local ruler and the coordinate cadence of its clock both scale with p. The clock's coordinate period therefore scales as p^{-1}; consequently, the coordinate speed of light defined by the ratio of one ruler interval to one clock period scales as $p/(p^{-1}) = p^2$.

Denoting the dimensionless pressure factor by p, this biconformal scaling is formalized as:

$$\frac{L_{\mathrm{coord}}}{L_0} = \frac{m_{\mathrm{eff}}}{m_0} = \frac{\Omega_t}{\Omega_{t,0}} = p, \qquad \frac{c_{\mathrm{coord}}}{c_0} = p^2, \qquad p = e^{\varphi/2}. \tag{5}$$

Here L_{coord} is the footprint of a local ruler in the distant asymptotic coordinate chart, not its locally measured proper length. For the local observer it remains one unchanged unit. The clock law gives

$d\tau = p\,dt$, while ruler calibration gives $d\ell = p^{-1}d\sigma$. The temporal metric coefficient is therefore p^2 and the spatial coefficient is p^{-2}. A null signal has coordinate speed p^2c_0, but reconstruction with the same local ruler and clock leaves the locally measured limiting speed exactly equal to c_0.

A similar power-law dependence appears historically in the Polarizable Vacuum (PV) models of Robert Dicke and Harold Puthoff [Dicke, 1957, Puthoff, 2002]. RefG relates these powers to the fundamental ontological distinction between nodes and the links connecting them.

Formally, the external macroscopic readout of a single oscillon couples to the foundation-medium pressure deficit through the exponential relation $e^{\varphi/2}$:

$$m_{\text{eff}} = m_0 e^{\varphi/2}. \tag{6}$$

On the deficit branch, $\varphi < 0$, confining the pressure factor to $0 < p < 1$. This relation describes the local external readout of an individual oscillon embedded within an inhomogeneous background: as φ deepens (pressure drops), its operational size and m_{eff} contract. Simultaneously, the total gravitational charge of the extended source generating the deficit is determined via a separate volumetric integration. This scaling follows directly from the theory's first-order biconformal branch.

Here, four physical quantities must be rigorously distinguished:

1. **Oscillon internal state and proper energy** (M_{proper});
2. **Single-oscillon effective refractive mass in a given background** (m_{eff});
3. **Active source profile of the induced pressure deficit** (M_{active});
4. **Total asymptotic gravitational charge of the system** (ADM/Komar mass: Arnowitt et al. 1962, Komar 1959).

These four quantities form two levels of accounting. The first distinguishes the oscillon's internal store from the share read through the surrounding medium. The second relates externally read mass to the active core and its foundation-medium dressing. At the asymptotic boundary these contributions combine into one charge, so that a single deficit trace is counted once.

An intuitive physical analogy captures this self-consistent dynamics: **imagine a crystal of colored ice placed in pure water.** The crystal is a locally frozen, ordered form of the same water, just as an oscillon is a localized resonant locking of the universal foundation-medium. As the crystal melts, its color spreads through the wider medium while the localized crystal becomes smaller. Here the spreading color represents redistribution from a localized node of organized structure into an extended trace in the medium.

The more widely the resonant trace is distributed, the more the local pressure, externally read mass, operational size, coordinate clock cadence, and coordinate speed of light change. Falling pressure and falling p weaken further resonant coupling and slow relaxation through negative feedback. On a homogeneous background this process approaches the fully relaxed boundary asymptotically.

This p/p^2 scaling is formalized through the biconformal metric:

$$ds^2 = e^{\varphi}c^2dt^2 - e^{-\varphi}d\boldsymbol{\sigma}^2, \qquad \varphi = -\frac{2U}{c^2}, \tag{7}$$

where $U > 0$ is the gravitational potential of the selected exponential branch and coincides at leading order with the magnitude of the Newtonian potential. In the weak-field limit ($U/c^2 \ll 1$), the metric components expand in a Taylor series:

$$g_{00} = e^{\varphi} \approx 1 - \frac{2U}{c^2} + \frac{2U^2}{c^4} + \cdots, \qquad g_{ij} = -e^{-\varphi}\delta_{ij} \approx -\left(1 + \frac{2U}{c^2} + \cdots\right)\delta_{ij}. \tag{8}$$

The common clock–ruler dictionary fixes the biconformal metric and its first-order spatial response. A static functional determines the profile around a material source through three explicit conditions:

the response is additive and logarithmic, $u = -\ln p$; the gradient term is canonical; and the coupling to the active source density entering Gauss's law is linear. Under these conditions, variation of the functional together with the asymptotic boundary and Gauss charge gives $u = GM/(c^2 r)$.

In this static, spherically symmetric weak-field description, the first post-Newtonian (1PN) metric expansion yields the Parameterized Post-Newtonian (PPN) coefficients:

$$\beta = 1, \qquad \gamma = 1. \tag{9}$$

Here $\gamma = 1$ is the first-order geometric consequence of the common clock–ruler response. The second-order logarithmic response gives $\beta = 1$. Thus γ follows directly from the common operational geometry, whereas β rests on the exact bridge between foundation microdynamics and the response $u = -\ln p$.

Relational Coframe, Einstein–Hilbert Geometry, and Full 1PN. The covariant equations in this subsection use the $(-+++)$ signature; the preceding biconformal presentation uses $(+---)$.

The static p profile displays the volumetric scalar projection of the full geometry. The complete state of the local geometry is represented by a **relational coframe** $e^A{}_\mu$—a system of local frame one-forms that combines volume, shape, shear, orientation, and the temporal direction. The coframe defines one operational metric,

$$g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu. \tag{10}$$

The spacetime orientation of the coframe and the oscillon's internal three-axis structure are physically distinct; color is associated with the latter in the particle map. Neighboring coframes are compared through a flat, metric-compatible inertial connection, while the electromagnetic channel has its own $U(1)$ connection. The failure of an infinitesimal parallelogram to close under coframe transport is **geometric torsion**—the measure of non-integrable deformation in the coframe continuum.

The leading low-energy class of the post-Genesis relational-coframe continuum is defined by three conditions:

1. The action is local, reversible, constant-coefficient, parity-even, and quadratic in torsion.
2. The phase sector is minimally coupled, with no unsuppressed mixed operator contributing at the same order.
3. The linear Minkowski limit is regular, while coframe orientation remains an inertial gauge freedom and does not generate an independent long-range physical orientation mode.

Within this class, the unique coefficient ratio gives the TEGR combination. The three independent quadratic torsion invariants are

$$I_1 = T^\rho{}_{\mu\nu} T_\rho{}^{\mu\nu}, \qquad I_2 = T^\rho{}_{\mu\nu} T^{\nu\mu}{}_\rho, \qquad I_3 = T_\mu T^\mu, \quad T_\mu = T^\nu{}_{\nu\mu}. \tag{11}$$

Their TEGR combination is

$$T_{\text{TEGR}} = \frac{1}{4} I_1 + \frac{1}{2} I_2 - I_3. \tag{12}$$

The selected low-energy action is therefore

$$S_F = -\frac{c_0^3}{16\pi G} \int d^4x\, e\, (T_{\text{TEGR}} + 2\Lambda_F) + S_C[g, J, \theta_C] + S_\partial, \tag{13}$$

where S_C is the covariant phase-current action and S_∂ is the matched boundary functional. The minus sign of the gravitational torsion term combines with the following torsion–curvature identity to recover the Einstein–Hilbert sign exactly. With the $(-+++)$ signature, the exact connection identity

$$R_{\text{LC}} = -T_{\text{TEGR}} + \frac{2}{e} \partial_\mu (e T^\mu), \tag{14}$$

converts the coframe action to the Einstein–Hilbert action up to a boundary term [Beltrán Jiménez and Dialektopoulos, 2020, Maluf, 2013]. Here $R_{\rm LC}$ is the Ricci scalar of the Levi–Civita connection, $e = \det(e^A{}_\mu)$ is the coframe determinant, and T^μ is the torsion trace. Equivalence at the action level is exact for compact-support variations with suitable asymptotic falloff, or with the exactly matched boundary functional. Through this identity, Einstein geometry is the continuum representation of the relational-coframe law. The Fierz–Pauli/Deser spin-2 bootstrap supplies an independent crosscheck of the same final operator.

S_C describes the phase current as a barotropic, isentropic, irrotational one-potential flow on its positive, stable, causal constitutive branch: $\rho_C > 0$, $\rho'_C > 0$, and $0 \le n_{C,\rm op}\rho''_C(n_{C,\rm op})/\rho'_C(n_{C,\rm op}) \le 1$. Variation with respect to θ_C gives the conserved current, whereas metric variation gives the Hilbert stress–energy tensor [Brown, 1993]

$$T^C_{\mu\nu} = (\rho_C + p_C)u_\mu u_\nu + p_C g_{\mu\nu}. \tag{15}$$

Here $n_{C,\rm op}$ is the phase-charge density per operational volume and $p_C = n_{C,\rm op}\rho'_C(n_{C,\rm op}) - \rho_C(n_{C,\rm op})$ is the thermodynamic pressure of the phase sector, whereas P_F is the operational readout of the foundation's background pressure. These are distinct physical variables; their constitutive relation belongs to foundation microphysics. With the source restricted to the phase-current sector, the field equation is

$$G_{\mu\nu} + \Lambda_F g_{\mu\nu} = \frac{8\pi G}{c_0^4} T^C_{\mu\nu}. \tag{16}$$

For the general case, the right-hand side contains $T^{\rm tot}_{\mu\nu}$; every physical stress–energy sector in the action enters that total once through its own metric variation.

The resulting action places RefG and General Relativity in the same 1PN equivalence class under five conditions:

1. Through 1PN order, the complete action consists of the Einstein–Hilbert operator and the matter action.
2. Matter couples universally and minimally to the same metric.
3. The source is recorded by one complete, conserved stress–energy tensor.
4. RefG and GR use the same local source, PPN gauge, background, and boundary conditions.
5. No additional unsuppressed field or operator participates in the 1PN degree-of-freedom inventory.

Under these conditions, the complete standard PPN vector is

$$\gamma = \beta = 1, \qquad \xi = \alpha_1 = \alpha_2 = \alpha_3 = \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0. \tag{17}$$

The result applies to isolated, weak-field, slowly moving, weakly self-gravitating sources. The static, spherically symmetric calculation provides an independent local check: $\gamma = 1$ follows from the common clock–ruler response, while $\beta = 1$ is tied to the logarithmic law $u = -\ln p$.

For a light-clock realization, the coordinate period is the ratio of coordinate ruler footprint to coordinate light speed ($\Delta t \sim L_{\rm coord}/c_{\rm coord}$). If a pressure drop halves that coordinate footprint ($p = 1/2$) while reducing coordinate light speed by a factor of four ($p^2 = 1/4$), the period doubles—the clock runs precisely twice as slow.

In the shared static, spherically symmetric regime, the 1PN effects controlled by $\beta = \gamma = 1$ recover the classical predictions of General Relativity: the gravitational deflection of light, Shapiro time delay, gravitational redshift, and planetary perihelion precession [Bertotti et al., 2003, Will, 2014]. In particular, the Cassini radio-link experiment measured $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$.

Historically, the additional **factor of two** arising from the spatial metric component g_{ij} was the decisive contribution that elevated Einstein's 1911 half-deflection (derived solely from temporal dilation) to the full geometric result of 1915–1916 [Einstein, 1916, Shapiro, 1964]. In RefG, this factor of two follows from two coordinate readouts of the same local standard: material size and clock cadence scale with p, while the coordinate speed of light reconstructed from them scales with p^2.

2.3 Refractive Attraction, Inertia, and the Absence of Friction

An oscillon, as an extended wave configuration, propagates through spatial gradients of foundation-medium pressure and coordinate wave speed. The phase velocity is higher on the denser side and lower on the rarefied (deficit) side. By virtue of the Huygens–Fresnel principle and the eikonal equation, the wavefront bends toward the region of lower pressure and reduced coordinate velocity—undergoing **refraction**. In the static weak-field limit, the pressure deficit generated by a massive body constructs an effective optical-geometric medium, and the refracted trajectory of a matter wavepacket appears macroscopically as gravitational attraction.

In RefG, Newton's falling apple is read as follows: the oscillon forms of the apple's atoms bend refractively toward the region of lower pressure and slower coordinate propagation established by the Earth. Familiar gravitational attraction is the direct local response of wave-like matter to an inhomogeneous medium.

In the relational-coframe description, the pressure deficit and non-integrable deformation describe the state of the gravitational geometry. The scalar p profile gives the common volumetric response of clocks, rulers, and propagation, while the coframe carries the full tensor structure of the geometry. In the post-Genesis continuum, the TEGR action equals the Einstein–Hilbert action up to a boundary term; metric variation of the phase-current sector yields the Hilbert source $T^{C}_{\mu\nu}$ on the right-hand side. Every physical stress–energy contribution present in the action enters the total Hilbert tensor once. A gravitational well is a region of depressed pressure and altered geometry, in which the total mass of a bound system is smaller than the sum of its separated constituents by the mass equivalent of the binding energy ($\Delta M = B/c^2$).

GW150914 provides a clear observational example of this energy balance: the difference between the initial binary mass and the final remnant mass—equivalent to roughly three solar masses ($3\,M_{\odot}$)—was carried away by gravitational waves [Abbott et al., 2016]. In the language of RefG, two oscillatory sources reorganize into a single, deeper deficit profile. In the TEGR-equivalent low-energy geometry, this rapid nonlinear change is read as the standard transverse-traceless tensor wave.

RefG associates **inertia** with the dynamic deformation of an oscillon's deficit profile during acceleration. When an oscillon is accelerated by an external force, its pressure profile develops a fore-aft asymmetry along the direction of motion; this gradient asymmetry generates a reactive resistance from the medium opposing the acceleration (the inertial force). Under deceleration, the profile reverses symmetrically.

Along a Lorentz-covariant, localized finite-energy branch satisfying the Laue condition ($\int T^{ij}\, d^3x = 0$ for all i, j), the inertial coefficient is determined by the total Noether energy of the system [Noether, 1918, von Laue, 1911]. Coupling this same energy as the universal source of gravity yields the equivalence $m_i = m_g$ at leading order. The MICROSCOPE satellite mission tests this universality at the 10^{-15} level [Touboul et al., 2022].

For an oscillon in uniform, rectilinear motion, the wave pattern is transferred symmetrically across foundation nodes, leaving no dissipative wake or net energy loss. RefG identifies **frictionless inertial motion** with this coherent phase-relay regime: motion is the wave-like relay of form. In the low-energy continuum description, the common Lorentzian cone relates the limiting rate of phase transmission to the metric light cone.

2.4 Local Unobservability and the Principle of Relativity

A local observer within a gravitational field reads time and spatial scale as normal because the observer's clock, ruler, and body are made from the same foundation-medium. When pressure changes, the entire local system is transposed by one common factor. A change slow relative to internal periods is adiabatic: it preserves the oscillatory mode and excites no quantum transition. On this branch, a dimensionless ratio is exactly invariant when both of its components follow the same adiabatic transposition; RefG places the fine-structure constant α in this class.

Two classes of experiment establish direct observational bounds on this picture:

- **Gravitational Redshift:** From the Pound–Rebka experiment [Pound and Rebka, 1960], through the Gravity Probe A hydrogen maser [Vessot et al., 1980], to the Galileo satellite tests [Delva et al., 2018, Herrmann et al., 2018], experiments measure with high precision the dependence of clock cadence on gravitational potential. RefG reads this dependence as the temporal channel of the common pressure–geometric response.
- **Invariance of Dimensionless Ratios:** Absorption spectra of distant quasars and laboratory optical atomic clocks constrain any cosmological variation of α to $|\Delta\alpha/\alpha| \lesssim 10^{-6}$ [Kotuš et al., 2017]. RefG reads this observational bound as a stringent test compatible with the prediction of collective-cadence self-calibration.

The null result of the Michelson–Morley experiment reflects the medium's **common-mode self-calibration**, whereas gravitational-wave detections by LIGO, Virgo, and KAGRA represent a **differential tensor response**:

- When the foundation-medium shifts uniformly and isotropically across a local domain, measuring rods, clocks, and optical propagation reconfigure synchronously in common mode, producing common-mode self-calibration in local measurements (modern optical cavities rule out spatial anisotropy in the speed of light to $\Delta c/c \lesssim 10^{-17}$; Herrmann et al. 2009).
- A gravitational wave, by contrast, is a time-dependent, quadrupolar differential strain $(h_+, h_\times)$: it stretches space along one transverse axis while compressing it along the perpendicular axis. This differential deformation breaks common-mode cancellation, imprinting a measurable phase shift in interferometer arms.

FitzGerald–Lorentz length contraction [FitzGerald, 1889, Lorentz, 1892] is the historical precursor to this mechanism. RefG attributes this contraction to the fact that measuring instruments are themselves wave configurations of the medium.

In the TEGR-equivalent low-energy geometry, tensor waves propagate on the common metric light cone, so $v_{\rm gw} = c$ at this order. The simultaneous detection of gravitational waves and a gamma-ray burst from the binary neutron star merger GW170817 tightly constrains the fractional difference between gravitational and electromagnetic propagation speeds [Abbott et al., 2017]:

$$-3 \times 10^{-15} \leq \frac{v_{\rm gw} - c}{c} \leq +7 \times 10^{-16} \tag{18}$$

Concurrently, the orbital decay of relativistic binary pulsars (PSR B1913+16, PSR J0737-3039A/B) confirms general-relativistic quadrupole radiation damping to an accuracy of 10^{-3}–10^{-4} [Weisberg and Huang, 2016, Kramer et al., 2021], while bounds on scalar breathing modes from the GWTC-3 catalog tightly constrain scalar wave channels [LIGO Scientific Collaboration et al., 2022].

With metric signature $(-+++)$, the clearest operational invariant in this picture is the difference between the proper times accumulated by two clocks along distinct gravitational and kinematic worldlines:

$$\tau_a = \frac{1}{c}\int_{\gamma_a} \sqrt{-g_{\mu\nu}dx^\mu dx^\nu}, \qquad \Delta\tau = \tau_1 - \tau_2. \tag{19}$$

The numbers displayed when the clock faces are reunited are the invariant result of physical histories traversing different pressure regimes, whereas the instantaneous cadence and size measured locally are relative readouts within the common medium.

2.5 Historical Context and the Role of RefG

The refractive, optical-medium reading of gravity possesses an illustrious heritage. In 1911–1912, prior to completing the geometric formulation of General Relativity, Albert Einstein related the coordinate speed of light to the gravitational potential [Einstein, 1911, 1912]:

$$c(\Phi) = c_0\left(1 + \frac{\Phi}{c^2}\right) \tag{20}$$

Medium optics has even earlier roots: in 1851, Hippolyte Fizeau demonstrated that moving water partially drags light (governed by Fresnel's dragging coefficient; Fizeau 1851), and in 1923, Walter Gordon proved that electromagnetic wave propagation in a moving dielectric medium obeys the null geodesics of an effective Lorentzian (pseudo-Riemannian) metric [Gordon, 1923], establishing the first formal bridge between continuum optics and spacetime geometry.

In 1957, Robert Dicke formulated gravitation through a variable vacuum refractive index [Dicke, 1957]. The modern discipline of **analogue gravity** was inaugurated by William Unruh in 1981, demonstrating that sound waves in transonic fluid flows propagate according to an effective acoustic metric exhibiting an event horizon [Unruh, 1981].

Grigory Volovik's condensed-matter program [Volovik, 2003] and the extensive review by Barceló, Liberati, and Visser [Barceló et al., 2011] established how local Lorentz invariance and curved effective geometries emerge from the microscopic dynamics of quantum condensates such as superfluid helium-3 and helium-4. A striking laboratory success followed in 2019, when Jeff Steinhauer's group experimentally observed the acoustic analogue of thermal Hawking radiation in a rubidium Bose–Einstein condensate [Muñoz de Nova et al., 2019].

Parallel to the medium-optics approach stands the **thermodynamic and entropic** school of gravity:

- Andrei Sakharov interpreted gravity as an elastic vacuum phenomenon induced by quantum fluctuations [Sakharov, 1967];
- Ted Jacobson demonstrated that Einstein's field equations arise directly as an equation of state from the thermodynamic entropy balance ($\delta Q = TdS$) across local Rindler horizons [Jacobson, 1995];
- Erik Verlinde formulated gravitational attraction as an entropic force driven by holographic information gradients [Verlinde, 2011].

These approaches and RefG share the view that spacetime geometry is a derived physical structure. RefG identifies its source with a concrete pressure–refractive foundation-medium.

Refractive Gravity (RefG) organizes this historical legacy into four unifying steps:

1. **Monistic Ontological Unity:** The foundation-medium carries the geometric response while generating the oscillon forms of matter, measuring instruments, and the observer. Hydrodynamic intuition thereby becomes part of one calculable physical architecture.
2. **Local Population-Locked Cadence (Ω_t):** Nonlinear synchronization provides a concrete mechanism for local self-calibration. For a physical oscillon, this mechanism predicts a spectrum in which every stable-mode frequency scales with the common pressure factor while dimensionless ratios remain invariant.
3. **One Foundation with a Multicomponent Geometric Response:** Pressure and expansion govern the scalar readout, symmetric shape and shear carry the tensor response, and orientation is an inertial gauge freedom. Together these components form one operational geometry, and every physical stress–energy contribution present in the action couples to its metric once.
4. **From Relational Coframe to Einstein–Hilbert:** In the post-Genesis coframe's leading local, reversible, constant-coefficient, parity-even quadratic-torsion class, with minimal phase coupling and no unsuppressed mixed operator at that order, the regular linear Minkowski limit determines the TEGR coefficient ratios by requiring the absence of an independent physical orientation mode. The exact torsion–curvature identity converts the action to Einstein–Hilbert form up to a boundary term [Beltrán Jiménez and Dialektopoulos, 2020, Maluf, 2013]. The Fierz–Pauli linear spin-2 construction and Deser's universal self-coupling provide a separate route to the same final operator. In four dimensions, Lovelock's theorem gives the corresponding uniqueness statement for local metric field equations of at most second differential order, up to normalization and a cosmological term [Deser, 1970, Lovelock, 1971].

In this historical landscape, RefG brings optical refraction, topological solitons, and thermodynamic balance into one research program, with a distinct mathematical and observational test for each link.

3 Three Scales and Cosmological Horizons

3.1 The Local Scale: Compact Objects, Internal State, and External Readout

The universal pressure chain constructed throughout the preceding chapters manifests across three distinct astrophysical regimes:

1. **Local scale:** the localized pressure-deficit profile surrounding oscillons and compact relativistic bodies;

2. **Galactic scale:** the coherent vortical flow of the foundation;

3. **Cosmological scale:** relaxation of the background pressure driven by geometric expansion of the foundation-medium, together with two complementary descriptions of time.

These are three dynamical regimes of one foundation. The local regime displays the mechanism in its sharpest form, the galactic regime reveals its coherent long-range response, and the cosmological regime describes the common historical evolution of the foundation.

In an extreme gravitational regime, an object's **internal state** (M_{proper}) and its **externally read active gravitational mass** (M_{external}) become physically distinct readouts. The compact-object map of RefG strictly separates the stored internal content of accumulated matter from the gravitational charge read at a distance. A remote observer reads the state through its pressure-deficit profile, clock cadence, and changing operational size, while the interior geometry, horizon, and global causal structure are fixed by the complete nonlinear field solution.

Two-channel accounting separates the object's local internal state from its asymptotic gravitational readout. The internal channel tracks the local state of matter, conserved charges, and proper scale; the external channel registers the deficit profile, clock cadence, operational size, and total mass read from afar. Gravitational compression drives these readouts sharply apart while preserving their relation to the same physical object.

RefG relates the mass defect of a merging compact system to nonlinear changes in binding and in the deficit profile. Its quantitative connection to the outgoing tensor flux is determined by the complete dynamical solution, with the gravitational-wave event GW150914 providing the experimental benchmark [Abbott et al., 2016].

The selected regular-core compact map permits a sharp separation between internal proper scale and external coordinate size. In this picture the object can appear compact, dark, and nearly "frozen" from outside while its proper geometry retains a different volumetric scale. The magnitude of that separation, the horizon, and the global causal structure are determined by the complete strong-field solution. In both descriptions the locally measured speed of light remains invariant (c); what changes is the temporal and spatial readout constructed by a distant observer. The "frozen star" is a regime of external readout, while the historical intuition of a "dark star" dates back to John Michell in 1784 [Michell, 1784].

Avoiding a singularity in this map requires a regular core: local density, curvature, and interior evolution must remain finite everywhere. One necessary premise of Roger Penrose's singularity theorem [Penrose, 1965] is the null convergence condition $R_{\mu\nu}k^{\mu}k^{\nu} \geq 0$. Through Einstein's equations, this is linked to the null energy condition (NEC), $T_{\mu\nu}k^{\mu}k^{\nu} \geq 0$. The effective source of the regular core violates the NEC in the required region. The stable phase current, for which $\rho_C + p_C > 0$, remains in an NEC-compatible sector; the compact core therefore requires an independent physical equation of state.

Regular compact objects represent an active frontier in relativistic astrophysics. A prominent example is the Mazur–Mottola "gravastar"—a vacuum condensate object devoid of a singular core [Mazur and Mottola, 2004]. Such alternatives can be empirically distinguished from classical Kerr black holes via their Event Horizon Telescope (EHT) shadow morphology [Event Horizon Telescope Collaboration, 2019], quasinormal mode ringing, and late-time gravitational-wave "echoes" [Cardoso and Pani, 2019]. RefG shares structural kinship with this family, anchoring its regular interior in the pressure-energy balance of the universal foundation-medium.

3.2 The Galactic Scale: Vortical Flow and MOND

Observational Reality. In disk galaxies, rotation curves at large radii do not follow the Keplerian decline expected from visible baryonic matter alone: they exhibit rich morphological variety, frequently flattening or declining far more gently than Newtonian predictions. Concurrently, baryonic mass and asymptotic rotation speed obey a tight empirical relation—the Baryonic Tully–Fisher Relation ($M_b \propto v_f^4$; Tully and Fisher 1977, Lelli et al. 2016). This observational program began with Vera Rubin and Kent Ford's measurement of the Andromeda galaxy [Rubin and Ford, 1970]. In the modern **Radial Acceleration Relation (RAR)**, 2,693 radial data points from 153 galaxies collapse onto a single universal curve [McGaugh et al., 2016].

The standard ΛCDM model accounts for these observations through approximately spheroidal cold-dark-matter halos. Modified Newtonian Dynamics (MOND), by contrast, introduces a universal acceleration threshold:

$$a_0 \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2}, \tag{21}$$

below which gravitational acceleration transitions from Newtonian a_N to the deep-MOND regime $\sqrt{a_N a_0}$ [Milgrom, 1983]. Relativistic completions (such as TeVeS and Skordis–Złośnik models) incorporate additional dynamical vector and scalar fields [Bekenstein, 2004, Skordis and Złośnik, 2021].

The RefG Vortical Mechanism and Foundation-Medium Polarization. At galactic scales, RefG relates the additional deficit to large-scale **vortical flow and induced polarization** of the foundation-medium. An ordered rotational regime deepens the central pressure deficit, while baryonic oscillons and the surrounding flow form one shared refractive profile.

At the macroscopic level, this response is described by a Ginzburg–Landau-type local effective potential:

$$W(g_v, g_N) = \frac{1}{3}g_v^3 + \frac{1}{2}g_N g_v^2 - a_0 g_N g_v, \tag{22}$$

where g_v is the foundation-medium response field, g_N is the baryonic Newtonian acceleration, and $a_0 \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2}$ is the universal transition acceleration scale. The variational equilibrium condition, $\partial W/\partial g_v = 0$, yields:

$$g \equiv g_N + g_v, \qquad g_N = \frac{g^2}{g + a_0}. \tag{23}$$

In the deep asymptotic regime ($g \ll a_0$), this relation approaches $g^2 \simeq a_0 g_N$, yielding for circular orbits the Baryonic Tully–Fisher Relation ($v^4 = GM_b a_0$) without individual galaxy tuning parameters.

Observational Check on SPARC Data. Distinct from the published 153-galaxy, 2,693-point RAR sample, the cuts $\Delta V/V < 10\%$ and $V_{\mathrm{obs}} > 20$ km/s select 2,788 points from 163 SPARC galaxies. With fixed a_0 and fixed mass-to-light ratios ($\Upsilon_{\mathrm{disk}} = 0.5$, $\Upsilon_{\mathrm{bulge}} = 0.7$), and without galaxy-by-galaxy parameter fitting, this comparison gives an RMS scatter of 0.141 dex and a mean residual of -0.012 dex (Figure 2).

Disk Geometry and QUMOND Numerical Test. For a finite-thickness Miyamoto–Nagai disk, transverse gradients of the algebraic field generate a geometric curl correction. On a 300×300 grid with $a = 3$ kpc and $b = 300$ pc, Biot–Savart reconstruction gives, in this finite-grid configuration, a maximum radial-acceleration correction of 9.23% relative to the one-dimensional algebraic mapping over $a \le r \le 5a$.

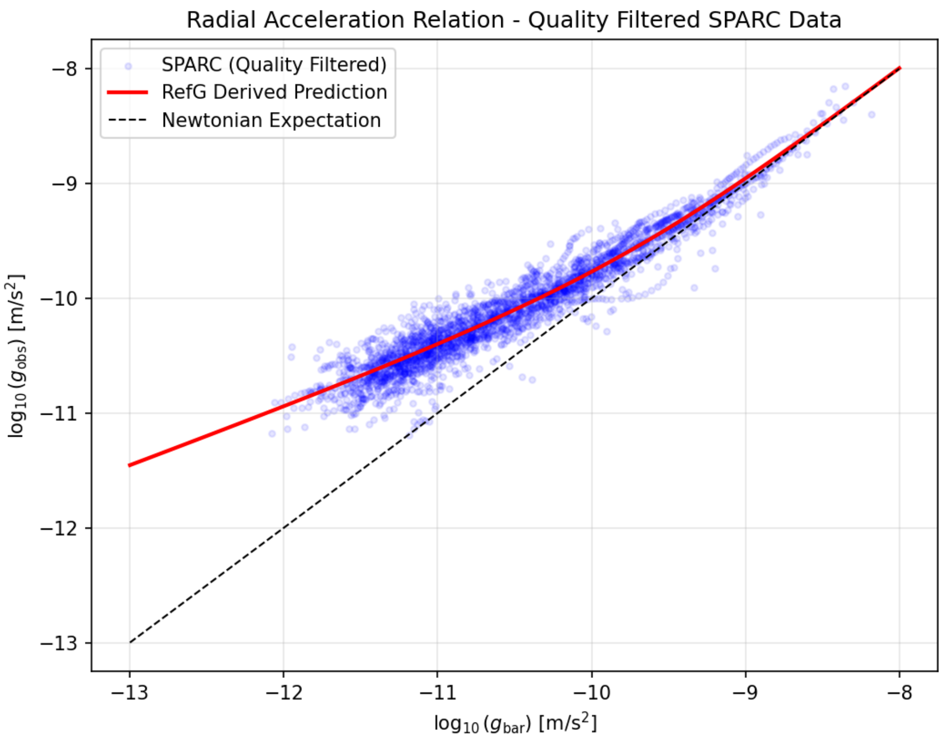

Figure 2: **Radial Acceleration Relation comparison with SPARC data.** The observed radial acceleration $g_{\rm obs} = V_{\rm obs}^2/R$ versus the baryonic acceleration $g_{\rm bar}$ for 2,788 selected points across 163 disk galaxies. With fixed a_0 and mass-to-light ratios, but no galaxy-specific fitting, the algebraic RefG curve obtained from $W(g_v, g_N)$ gives an RMS scatter of 0.141 dex and a mean residual of -0.012 dex.

Refractive Gravitational Lensing. Along the flat rotation curve branch, $\Phi_{\text{eff}} = v_f^2 \ln(r/r_0)$ and $n(r) \approx 1 - 2\Phi_{\text{eff}}/c^2$. Integrating the transverse refractive index gradient yields a deflection angle $\hat{\alpha} = 2\pi v_f^2/c^2$, matching the prediction of General Relativity for a singular isothermal sphere. Rotation and lensing thus emerge as two readouts of a single effective potential.

The closest modern analogue is the **Superfluid Dark Matter** model of Berezhiani and Khoury, in which a MOND-like force arises from phonon excitations of a condensate [Berezhiani and Khoury, 2015]. Both approaches draw on the collective response of a coherent medium. In the superfluid-dark-matter model that medium is the phase of an additional particle component; in RefG, the vortical flow is a large-scale regime of the unified foundation-medium itself.

Morphology. In RefG, galactic morphology reflects the flow regime of the foundation: ordered laminar rotation is associated with disk and spiral structure, whereas more chaotic or isotropic flow is associated with dispersion-supported elliptical galaxies.

Galaxy Clusters. In galaxy clusters, modifying the acceleration law alone does not fully resolve the dynamical mass deficit (historically highlighted by Fritz Zwicky in the Coma cluster; Zwicky 1933). RefG interprets the cluster mass budget through three distinct channels:

1. The global nodal deficit of the long-wavelength resonant network;
2. Individual galaxies and the long-range pressure-deficit fields of their constituent oscillons;
3. Cluster-scale vortical circulation and the hydrodynamic memory of past mergers.

All three enter a unified pressure-deficit budget, each counted exactly once.

Two decisive observational signatures distinguish this framework:

1. **Quantized Circulation:** If the foundation flow possesses a single-valued condensate phase, circulation is quantized in units of:

$$\Gamma = \oint \mathbf{v} \cdot d\mathbf{l} = n\frac{h}{m}, \tag{24}$$

   where m is the effective mass of the flow mode. For ultralight modes ($m \sim 10^{-22}$ eV), vortex filaments could be detected via weak lensing or stellar disk kinematics [Hu et al., 2000, Hui et al., 2021].

2. **The Frozen Memory of the Bullet Cluster:** In the Bullet Cluster (1E 0657-558), weak-lensing centroids are displaced from the X-ray gas at 8σ significance and follow the collisionless galaxies [Clowe et al., 2006]. RefG reads this separation as the **frozen memory of vortical flow in the foundation-medium**: during the collision, hydrodynamic drag slowed the collisional gas, while the deficit trace retained its motion with the galaxies for longer. The memory timescale, spatial profile, and lensing amplitude belong to one dynamical picture.

The galactic mechanism is jointly testable against the SPARC/RAR relation [Lelli et al., 2016], galaxy-cluster mass budgets, and Bullet-Cluster-type mergers. A combined test with the CMB, BAO, and large-scale-structure statistics will determine whether this galactic response and the cosmological neutral collective-phase source share one foundation-level normalization.

3.3 The Cosmological Scale: Foundation Expansion and Its Internal Readout

The cosmological history of RefG begins with the centerless, foundation-wide transition described in Section 1.2. It is the common boundary of the first distinguishable relations. Distance and direction acquire physical meaning only after the first distinguishable connections have formed. In the centerless initial state, equivalent regions share one common boundary of birth; a history ordered outward from a privileged site would break that initial symmetry. Local waves and oscillon centers arise only within the spatial layer that follows.

The sequence is: formation of the first relations, localization of oscillon seeds, overlap of their resonant fields, emergence of resonant coupling as the distinguishability threshold is crossed, growth of network connectivity, formation of stable oscillons, and long-term foundation expansion accompanied by pressure relaxation.

Four pressure variables carry distinct physical roles in this cosmological picture. The foundation background pressure P_F is an operational readout that controls the common response factor p of material scale and clock cadence. The quantity $p_C = n_{C,\mathrm{op}}\rho'_C(n_{C,\mathrm{op}}) - \rho_C(n_{C,\mathrm{op}})$ is the thermodynamic pressure of the neutral phase current. The ordinary thermal pressure is denoted by P_{th}, while P_* is the mechanical response of a homogeneous stationary vacuum phase.

The corresponding equations of state relate these four quantities. Keeping them distinct assigns every physical contribution of energy and pressure to its proper sector and prevents double counting. The geometric contribution Λ_{phase} of a constant phase density enters the total $\Lambda_{\mathrm{eff}} = \Lambda_{\mathrm{bare}} + \Lambda_{\mathrm{phase}}$ once. The initial thermal state is determined by the sectoral distribution of energy, the equation of state for P_{th}, and energy exchange with radiation.

The observable physical content of the thermal state is encoded in dimensionless ratios, including

$$\Theta_\gamma = \frac{k_B T_\gamma}{E_{\mathrm{atom}}}, \qquad \mathcal{R}_i = \frac{\Gamma_i}{H_A}. \tag{25}$$

Here E_{atom} is the atomic energy scale, Γ_i is the rate of the relevant microphysical process, and $H_A = (1/A)dA/d\tau$ is the expansion rate of the operational scale in process time. On the selected standard low-energy operational branch, the common p scale is already encoded in the metric and does not re-enter these dimensionless ratios as a second factor; the observable content of temperature is determined by the actual energetic relation between matter and radiation. The CMB spectrum, recombination, and BBN provide a joint test of this thermodynamic map.

The quantity $R_{\mathrm{act}}(t)$ denotes a local post-origin activation or correlation scale within which nodal relations reach the minimum distinguishability threshold ℓ_*. It is a local quantity; the homogeneous global geometry of the Universe is described by A. On a homogeneous background the same local rule applies around every node comoving with the network.

On the selected cosmological branch, formed matter follows the expanding nodal lattice of the foundation. For two galaxies comoving with the mean flow, let N_{12} denote the number of inter-node intervals along their comoving separation. In the ideal expansion component, N_{12} remains fixed while the physical length of each interval, $\ell_F(t)$, grows:

$$L_{12}(t) = N_{12}\,\ell_F(t). \tag{26}$$

The constancy of N_{12} applies specifically to the cosmological expansion component. Matter can move locally with respect to the lattice, rotate, collide, and undergo gravitational displacement; those processes can change N_{12}. In the selected ontological picture, growth of foundation separation in the mean Hubble flow is carried by the changing inter-node length scale.

The observer, the clock, and the ruler are made from the same foundation. The change in foundation scale is therefore read internally through the pressure response of the material standard. Define

$$a(t) = \frac{\ell_F(t)}{\ell_{F,0}}, \tag{27}$$

as the inter-node length scale of the foundation, and let $p(t)$ denote the material-standard scale. The causal chain of RefG is

$$a\uparrow \longrightarrow\ P_F\downarrow \longrightarrow\ p\downarrow. \tag{28}$$

The operational ratio available to an internal observer is

$$A(t) = \frac{a(t)}{p(t)}. \tag{29}$$

The ratio A is the internal operational readout; a and p are not measured directly and separately. The internal activation scale $R_{\mathrm{act}}(t)$, the foundation length scale $a(t)$, and the material standard $p(t)$ are three distinct quantities. The first describes the local spread of stably distinguishable relations, the second the change in foundation inter-node length, and the third the pressure response of material size and cadence.

Neutral Collective Phase and the Relaxation Law. A concrete post-Genesis realization of the relaxation law describes the organized foundation state by a neutral collective phase. In the foundation-volume representation,

$$\Psi_C = \sqrt{n_{C,F}}\, e^{i\theta_C}. \tag{30}$$

Here $n_{C,F}$ is the organized collective phase-charge density per foundation volume, while θ_C is its independent internal coordinate. The density of the same conserved charge per operational volume is denoted by $n_{C,\mathrm{op}}$. Node number, energy, electric charge, process time, and the periodic phase of an oscillon belong to different physical roles and coordinates of the foundation.

The common shift symmetry of θ_C generates a local current and conserves its total Noether phase charge Q_C within a fixed comoving region. Define the normalized foundation-density mean by $\eta_F(t) \equiv \langle n_{C,F}\rangle / \langle n_{C,F}\rangle_0$. On a homogeneous, isotropic background comoving with the network, the directional current vanishes. As the three-dimensional volume grows, the same Q_C is distributed over a larger measure, so η_F decreases:

$$\frac{Q_C}{Q_{C,0}} = \eta_F a^3 = 1. \tag{31}$$

A covariant phase-current action connects this conservation law to the post-Genesis operational geometry. With the $(-+++)$ signature, variation of θ_C gives $\nabla_\mu J^\mu = 0$, while metric variation gives the Hilbert stress–energy tensor,

$$p_C = n_{C,\mathrm{op}}\rho_C'(n_{C,\mathrm{op}}) - \rho_C(n_{C,\mathrm{op}}), \qquad T^C_{\mu\nu} = (\rho_C + p_C)u_\mu u_\nu + p_C g_{\mu\nu}. \tag{32}$$

This action describes a barotropic, isentropic, irrotational subfamily governed by one phase potential. On the homogeneous background, the covariant current gives $n_{C,\mathrm{op}}A^3 = \mathrm{const}$. The exact Jacobian $n_{C,\mathrm{op}} = p^3 n_{C,F}$, together with $A = a/p$, reads the same conservation law in foundation measure as $\eta_F a^3 = 1$. These are two volume representations of one charge Q_C, not two substances. The oscillon microphysics enters through $\rho_C(n_{C,\mathrm{op}})$, while the constitutive relation between p_C and the foundation background pressure P_F turns phase-current conservation into the background relaxation law.

On the selected homogeneous, isotropic branch comoving with the network, the constitutive relations set $P_F/P_{F,0} = \eta_F$ and $p^2 = \eta_F$. Together with phase-current conservation, they give

$$\frac{P_F}{P_{F,0}} = a^{-3}, \qquad p = \sqrt{\frac{P_F}{P_{F,0}}} = a^{-3/2}, \qquad A = \frac{a}{p} = a^{5/2}. \tag{33}$$

On this homogeneous branch, η_F, P_F, and p follow one redistribution of a conserved phase charge. As the three-dimensional foundation scale grows, the same Q_C is spread over a larger volume, the background pressure falls, and the material standard settles into its new state.

The event is one: the inter-node scale of the foundation grows; that growth lowers the background pressure, and the pressure decline coherently reorganizes the operational size of matter, its externally read mass, and its process cadence. Foundation separation and the material standard are not measured directly as separate quantities. An internal observer reads their ratio through $A = a/p$. Pressure relaxation and the reduction of the material standard are therefore the internal physical trace of foundation expansion, not independent cosmological effects to be added together.

Two independent conserved records operate in this picture. For two ideal-comoving material systems, the registered number of steps N_{12} along a selected relational path remains fixed even while the physical scale carried by each step changes. The quantity N_{12} records the structure of the selected path, whereas

Q_C is the Noether charge of the neutral collective phase. This separation preserves the distinct roles of cosmological scale kinematics and phase-source dynamics.

On the selected homogeneous branch, knowledge of A uniquely reconstructs the normalized foundation scale, the material standard, and the background-pressure state:

$$a = A^{2/5}, \qquad p = A^{-3/5}, \qquad \frac{P_F}{P_{F,0}} = \eta_F = A^{-6/5}. \tag{34}$$

Within this branch, it is a universal relative standard. It reads one measured cosmological history as the growth of the foundation scale, pressure relaxation, and the coherent response of the material standard. The universal measure is the unified network of ratios itself.

On the selected constitutive branch, a local strong field and the cosmological background display the same pressure–standard response. For the mass of Earth, the Schwarzschild diameter is approximately 1.8 cm—about the size of a hazelnut—and illustrates the extreme geometric scale of external compactness. The exact relation between that geometric scale and the pressure–standard response of RefG is determined by the complete strong-field solution. On the cosmological background the pressure–standard response acts continuously and universally. Our bodies, clocks, and rulers participate in the same process and continue to read their own units as locally unchanged.

Refractive Geometry and Background Dynamics. On the selected post-Genesis relational-coframe branch, and in the homogeneous description restricted to the phase-current source, the action is equivalent to Einstein–Hilbert up to a boundary term; variation of the induced metric gives

$$G_{\mu\nu} + \Lambda_F g_{\mu\nu} = \frac{8\pi G}{c_0^4} T^C_{\mu\nu}. \tag{35}$$

Here $T^C_{\mu\nu}$ comes from metric variation of the phase-current sector, while variation of θ_C gives current conservation. When additional matter and radiation sectors are included, the right-hand side contains the total $T^{\mathrm{tot}}_{\mu\nu}$. On the spatially flat, homogeneous, and isotropic branch, the operational A geometry takes Friedmann form, and the phase-current action determines its conserved stress–energy source. The complete right-hand side is the sum of the Hilbert tensors of all localized, material, and radiative sectors; each sector couples once to the common metric. Writing the already combined effective-action coefficient as $\Lambda_F \equiv \Lambda_{\mathrm{eff}}$ leaves one vacuum slot on the geometric side; when positive, it produces late operational acceleration.

The relations $A = a^{5/2}$ and $d\tau = p\,dt$ display the same cosmological history in two languages: a follows the inter-node scale of the foundation, whereas τ is time read by material processes. Redshift, the kinematic Hubble rate, signal time dilation, and geometric distances all arise from the single A geometry. On the ideal homogeneous background, these observables form one operational equivalence class with spatially flat ΛCDM at equal background parameters.

The Cosmic Microwave Background and the Einstein–Boltzmann Continuation. The cosmic microwave background (CMB) brings the geometry of the Universe, small density perturbations, photon propagation, and the formation of neutral atoms into a single observable history. In RefG, that history is carried by the same operational metric g_{op} and its scale A. During the CMB epoch, the complete Einstein source registry is

$$G_{\mu\nu} + \Lambda_{\mathrm{eff}} g_{\mu\nu} = \frac{8\pi G}{c_0^4} \left(T^{be}_{\mu\nu} + T^{\gamma}_{\mu\nu} + T^{\nu}_{\mu\nu} + T^{C}_{\mu\nu} \right). \tag{36}$$

Here $T^{be}_{\mu\nu}$ describes the baryon–electron plasma, $T^{\gamma}_{\mu\nu}$ the photons, $T^{\nu}_{\mu\nu}$ the neutrinos, and $T^{C}_{\mu\nu}$ the neutral collective phase current. Each physical source enters once, while the already combined coefficient Λ_{eff} occupies one vacuum slot on the geometric side.

One conserved phase charge Q_C is read with two volume measures. On the homogeneous background, its density per foundation volume and its density per operational volume are related by the exact normalized Jacobian

$$\widehat{n}_{C,\mathrm{op}} = p^3 \widehat{n}_{C,F} = A^{-3}. \tag{37}$$

Here $\widehat{n}_{C,F} \equiv n_{C,F}/n_{C,F,0}$ and $\widehat{n}_{C,\mathrm{op}} \equiv n_{C,\mathrm{op}}/n_{C,\mathrm{op},0}$ are the corresponding normalized densities, with $A(t_0) = a(t_0) = p(t_0) = 1$. This exact relation belongs to the homogeneous background. CMB perturbations evolve directly through the operational covariant current; no independent local foundation metric or second substance is introduced.

On the CMB cold branch with fixed energy per conserved charge, each conserved unit has the constant positive energy μ_C, so that

$$\rho_C = \mu_C n_{C,\mathrm{op}}, \qquad p_C = 0, \qquad c_{s,C}^2 = 0, \qquad \sigma_C = 0. \tag{38}$$

Under these conditions, the background and linear scalar dynamics of $T^C_{\mu\nu}$ are exactly those of a pressureless, irrotational Einstein source. Its quantitative normalization Ω_{C0} remains an input to the subsequent CMB forward calculation; this closure does not derive its numerical value from the foundation. In a standard Boltzmann code, the `cdm` slot may be used as a computational alias for the same $T^C_{\mu\nu}$: the equations are retained, while the physical source is the collective phase current of RefG, with no separate particle cold source added beside it.

On the selected low-energy operational branch, standard matter, the Maxwell field, neutrinos, and local atomic/QED physics couple minimally to the same metric, while the local dimensionless atomic and QED ratios retain their standard values. The source $T^C_{\mu\nu}$ couples to the plasma only gravitationally, with no direct energy–momentum exchange. Within the linear Thomson collision kernel, the photon and baryon momentum transfers cancel exactly; the neutrino phase-space hierarchy retains its standard anisotropic-stress sector; and recombination follows the established atomic kernel. In operational conformal time and units with $c_0 = 1$, $T_\gamma A = \mathrm{const}$, the positive Thomson scattering rate is $\dot{\kappa} = A n_e \sigma_T > 0$, and $\kappa(\eta) = \int_\eta^{\eta_0} \dot{\kappa}\, d\bar{\eta}$. The common p scale is already represented by the operational metric and does not re-enter the atomic equations as a second factor.

With the standard unit-normalized adiabatic initial mode, the background evolution, linear scalar perturbations, photon–baryon collisions, neutrino hierarchy, recombination, and line-of-sight initial-value problem take exactly the standard Einstein–Boltzmann form. Once the same amplitude and scale dependence of the primordial $P_{\mathcal{R}}(k)$, the same reionization history, and the same remaining late-time inputs are supplied, numerical evolution of the inherited system yields the corresponding standard linear CMB transfer functions and unlensed angular spectra. RefG continues Einstein's cosmological dynamics and extends the physical interpretation of that established calculation by grounding the cold source in a conserved collective phase of the foundation.

The Time Bridge and a Finite Process History. Process time counts accumulated material cycles, whereas foundation coordinate time follows the underlying order of events. Their bridge is $d\tau = p\, dt$: the factor p specifies how many material cycles fit within one coordinate-time interval. Both measures begin with the first distinguishable relation, at $t = 0$ and $\tau = 0$.

In the early Universe, a large p places many internal cycles inside a short coordinate interval. When p is integrable, that abundance of cycles remains a finite process history. The Universe therefore has a beginning in both time measures, although coordinate and process clocks read its age through different cyclic standards. Each material structure begins its own process history at the epoch of its formation.

Supernova Time Dilation. In the cosmological picture of RefG, a larger early p means that material processes completed more cycles within the same interval of coordinate time than they do in the present background; the local τ standard serves as the normal clock within its own epoch. On the ideal transparent photon branch, spectroscopic redshift follows directly from the metric and from comparison of the source and receiver's local atomic standards:

$$\frac{A_0}{A_e} = \frac{a_0/p_0}{a_e/p_e} = 1 + z. \tag{39}$$

Neighboring null signals in the same metric independently show that an event lasting $\Delta\tau_e$ in source process time is recorded by the present observer with duration

$$\Delta\tau_{\rm obs} = \frac{A_0}{A_e}\,\Delta\tau_e = (1+z)\,\Delta\tau_e. \tag{40}$$

Type Ia supernova light curves and spectral aging exhibit precisely this $(1+z)$ time dilation [Blondin et al., 2008]. In RefG, expansion of the foundation scale a lowers P_F, the pressure response changes p, and the entire history from source to receiver is read through the single operational geometry $A = a/p$.

Independent calculations in the A language of RefG and the z language of spatially flat ΛCDM give the same ideal redshift, time dilation, Hubble function, angular-diameter distance, and luminosity distance at equal background parameters. For these observables, the two descriptions belong to one operational equivalence class. The distinguishing physics lies in source and clock microphysics, calibration, and inhomogeneous dynamics.

Early Structural Maturity (JWST). A supernova time profile compares clocks from different epochs through photon propagation. The maturity of a galaxy records accumulated process history: mass assembly, successive stellar generations, chemical enrichment, and dynamical organization are the products of many internal cycles. A larger early p may provide more accumulated process time. An independent model of collapse, cooling, star formation, and feedback determines how that additional process time is converted into galactic growth. The early massive galaxies observed by JWST are a stringent test of this branch [Labbé et al., 2023].

Two Observational Channels for One Mechanism. Supernova time dilation and the maturity of early galaxies provide two distinct observational channels. The first measures photon propagation and signal-arrival intervals; the second probes the structural consequences of accumulated process time. Together they connect global relaxation of the foundation, the historical evolution of clocks, and the growth of matter.

Ontological Consistency and the Arrow of Time. Within a single-medium ontology, the familiar **vacuum-energy problem** becomes a problem of common energy accounting. If a manifested vacuum phase carries a constant operational energy density ϵ_*, it produces the negative mechanical response $P_* = -\epsilon_*$ and $w_* = -1$ as the volume grows. Its geometric contribution is $\Lambda_{\rm phase}$, and the operational geometry depends on one effective cosmological term, $\Lambda_{\rm eff} = \Lambda_{\rm bare} + \Lambda_{\rm phase}$. Each physical contribution enters this total once through its own variational source. A fixed total energy stored in a comoving cell is diluted by expansion and has matter-like behavior, $w = 0$. The conserved quantity Q_C belongs to the phase-action sector, whereas the vacuum term belongs to the energy sector described by ϵ_*. The q-theory program provides a historical motivation for this mechanism [Volovik, 2003, Klinkhamer and Volovik, 2008].

RefG relates the macroscopic **arrow of time** to the global direction of foundation relaxation. Like a colored crystal dissolving in water, localized energy spreads across the many degrees of freedom of the medium and creates a statistically irreversible history. This mechanism can connect the thermodynamic, cosmological, and radiative arrows to one common source [Planck, 1899]. Its quantitative form is determined by the entropy law and the irreversible dynamics of the foundation.

3.4 Quantum Randomness and Two-Level Time

The two-level nature of time opens a new interpretive perspective on quantum physics. The place and moment of a measurable event are recorded in oscillon-clock time, while RefG associates the selection of a particular outcome with the internal level of the foundation-medium. Here "prior" denotes logical priority: clocks, distance, and the measurement procedure are structures that arise from this level.

Bell's theorem and its experimental tests rule out local hidden-variable models that explain outcomes through preassigned local properties [Bell, 1964, Hensen et al., 2015]. At the same time, the **no-signaling theorem** establishes that entanglement correlations cannot be converted into controllable superluminal communication.

RefG reads this phenomenon as **atemporal internal coordination**. At the foundation level, the correlation belongs to the coherent unity of a single state; in metric language it appears as a nonlocal statistical correlation between spacelike-separated measurements. The physical signal channel that carries information through space remains governed by local causality.

John Wheeler's **delayed-choice experiment** is read in the same language [Wheeler, 1978, Jacques et al., 2007]: in a single-photon interferometer, a late choice of measurement arrangement determines whether wave interference or which-path information is manifested. RefG describes this unity through the internal integrity of the phase state while preserving the metric order of events.

Related concepts in modern physics include:

- The atemporal ground level of the **Wheeler–DeWitt equation** ($\hat{H}\Psi = 0$; DeWitt 1967);
- The **Page–Wootters mechanism**, where time emerges from subsystem quantum correlations [Page and Wootters, 1983];
- **de Broglie–Bohm pilot-wave theory**, characterized by non-local phase coordination [Bohm, 1952].

Any such account is simultaneously constrained by **Born's probability rule** and **no-signaling**: it must reproduce the observed outcome statistics $P = |\psi|^2$ [Born, 1926] without permitting controllable superluminal communication.

4 Oscillons and the Origin of Matter

4.1 Many Patterns in One Ocean

To build physical intuition, imagine a single continuous ocean and the diverse wave motions that arise within it. Most waves form, propagate across the surface, and quickly dissipate; under specific nonlinear conditions, however, certain configurations self-organize into stable, localized, self-sustaining structures:

- One may form a toroidal vortex ring, with fluid circulating continuously in a closed loop;
- Another may manifest as a localized spherical pulsation (oscillon);
- A third may take shape as a helical, twisting wavepacket.

The particle ontology of RefG rests on one clear idea: **electrons, protons, photons, and quarks are distinct resonant and topological patterns of one universal foundation-medium.** A massive particle is a locally self-confined form of the medium; a photon is a freely propagating transverse mode. The diversity of matter arises from the different stable motions of one foundation.

A historical precursor is Tony Skyrme's model, in which nucleons are described as topological solitons—"skyrmions"—of a nonlinear meson field [Skyrme, 1961]. RefG extends this intuition to a common ontological foundation: every solitonic and propagating mode belongs to the dynamics of one medium.

The spatial picture of matter is continuous and volumetric-wave-like. The vibrational amplitude and phase of the foundation are distributed through space, while a discrete event in a detector is the resonant registration of that vibration by the detector's atomic oscillons. The amplitude and phase of the wave function thereby acquire a direct physical carrier.

In the RefG map, the atomic "electron cloud" is a real extended phase state. The measurement point arises through local locking with the detector, while the stationary atomic state is a standing-wave mode of the foundation.

The single-electron double-slit experiment displays both sides of this picture at once: the extended amplitude spans the geometry of both slits, while each registration on the screen appears as a discrete point. Akira Tonomura's group directly recorded the gradual accumulation of this statistical pattern [Tonomura et al., 1989]. A stationary atomic mode preserves its energy; radiation accompanies transitions between states.

The spatial spread of a wavepacket and the intrinsic structural form factor of a particle are distinct physical concepts:

- Elastic scattering probes charge and energy distribution form factors, while deep inelastic scattering reveals internal parton structure and distribution functions;
- The spatial extent of a propagating wavepacket does not enter these observables as an intrinsic physical size.

Empirically, the electron exhibits a pointlike electromagnetic form factor down to scales of order 10^{-18} m [Navas et al., 2024]. A direct empirical test of the oscillon framework is whether the form factor computed from its internal topological charge profile reproduces this pointlike behavior across accessible momentum transfer scales.

4.2 Localized Self-Confinement: Massive and Massless Modes

In the RefG particle framework, the defining ontological dividing line between massive and massless sectors is **localized self-confinement**.

Massive Oscillons. The energy of a massive particle is confined within a localized, self-sustaining configuration. When a wavepacket is trapped via vortex circulation or stable nonlinear pulsation within a finite spatial region, Bernoulli-type dynamics induce a persistent static pressure deficit in the surrounding medium.

Long-lived localized nonlinear pulsations are well-known in classical field theory as **pulsons and oscillons** [Bogolyubsky and Makhankov, 1976, Gleiser, 1994, Copeland et al., 1995]. RefG distinguishes two levels of protection: the derived phase-supported core is stabilized by an ordinary Noether charge, while topological and population-level locking provide the broader mechanism for particle-species selection and long-term stability. In external readout, the energy held in the localized structure manifests as inertia and effective gravitational mass.

The first concrete dynamical realization of this direction is built from a complex phase field minimally coupled to the coframe. Its nonlinear action produces a finite-energy, radially nodeless Q-ball-type core with an analytically determined existence window and convergent numerical evidence of orbital stability on the fixed Minkowski coframe. Population-level phase locking embeds this local protection within a broader mechanism that couples the core to the common environment and selects stable particle species.

Massless Carriers (The Photon). A massless mode possesses neither a rest frame nor a localized static core—it propagates through the network at the limiting phase transmission velocity (c). RefG interprets the photon as a freely propagating transverse phase mode.

Helicity and localized rotation are fundamentally distinct phenomena:

- The ± 1 helicity of a photon describes the handedness of its transverse polarization relative to the propagation axis, without forming a localized static vortex;
- A massive mode possesses a localized topological core and a static pressure-deficit well.

In Einstein–Hilbert geometry, every localized or radiative sector whose diffeomorphism-invariant action depends universally and minimally on the common metric enters the total gravitational source once through its Hilbert stress–energy tensor. The standard electromagnetic action includes photon energy and momentum in that source. A propagating photon carries energy–momentum; a permanent localized well belongs to a bound state. The same accounting includes the entire internal energy of a composite particle.

The first exact bridge to a quantum description of the oscillon field is now in place. Its amplitude χ and phase θ_O combine into the complex field $\Psi_O = (\chi/\sqrt{2})e^{i\theta_O}$. The full nonlinear covariant action retains the binding terms required for localization, while the quadratic part of that same action about the strict vacuum takes exactly the form of the free complex Klein–Gordon action. On the fixed Minkowski branch, standard canonical quantization of this quadratic sector gives its free vacuum excitations a free Hamiltonian bounded below and nonnegative after vacuum normalization, particle and antiparticle sectors of Fock space, a conserved global $U(1)$ phase charge, microcausality—the commutator vanishes at spacelike separation—a simple propagator pole on the mass shell $k_\mu k^\mu = -m^2$, and spin zero [Tong, 2006, Wigner, 1939]. In the low-energy limit, the same equation reduces to Schrödinger dynamics. Einstein–Hilbert geometry and standard free quantum matter are therefore carried by the same RefG coframe.

The quartic and sextic terms form the interacting binding sector of the localized core. A free spin-zero quantum is a small vacuum excitation, whereas the localized core is an interacting solitonic state. The resulting $U(1)$ charge is the conserved global phase charge of the oscillon field.

The free-vacuum O-sector is represented classically by $T_O^{(2)}$ and quantum mechanically by $:\widehat{T}_O^{(2)}:$; energy accounting uses the corresponding description appropriate to the classical or quantum regime. In a semiclassical Einstein equation, the quantum source is the renormalized expectation value of the stress–energy tensor, while the full nonlinear tensor also contains operators generated by the binding interactions.

4.3 Self-Forming Resonators and the Two-Level Filter

Nonlinear wave equations admit an infinite diversity of mathematical solutions; nature, however, realizes a sparse, strictly defined spectrum of stable elementary particles. Why does nature select this specific discrete set?

RefG organizes this selection mechanism through a **two-level dynamical filter**:

1. Level One: Kinematic and Population Filter. An acoustic handpan provides an intuitive analogy. While the resonant length of a guitar string is externally fixed by its wooden body, a three-dimensional oscillon must sculpt its own resonant cavity within the medium. On a handpan's surface, a stable acoustic note unites the fundamental frequency, its octave, and the compound fifth into a harmonic $1:2:3$ triad.

This triad is a structural rhythm: a self-forming resonator binds several mutually coherent modes into a single physical entity. In RefG, an oscillon's pressure-deficit trace changes the local geometry, and the changed geometry selects viable modes: a mode must propagate through the foundation while also locking into resonant coexistence with the surrounding particle population.

2. Level Two: Energetic Stability Filter. Any form that passes the kinematic filter but has an energetically accessible decay channel rapidly decays into lower-energy stable modes.

Hadronic resonances (Δ, ρ, ω) illustrate this: with typical lifetimes of 10^{-22}–10^{-24} s, they appear in the spectrum, decay rapidly, and redistribute their energy among lower-energy decay products, including pions and protons.

The observed stability of the electron and the channel-dependent experimental lower bounds on proton decay ($\tau/B \gtrsim 10^{34}$ years for leading channels) impose stringent limits on any topological protection mechanism. These bounds set the quantitative scale for the protection mechanism that RefG associates with conserved charges and population-level phase locking.

The mathematical language governing these filters is that of nonlinear **stable modes and invariant phase-space structures**—configurations that retain their form under small perturbations.

A stable particle is a self-consistent equilibrium: the wave configuration creates a pressure deficit in the medium, and that deficit feeds back to select and reinforce the configuration itself. In classical field theory, Sidney Coleman's Q-balls are protected by conserved internal Noether charges [Coleman, 1985]. The local protection of the derived core rests on precisely such a Noether charge; population-level locking describes its coupling to the common environment and the selection of particle species.

This collective synchronization requires no external coordinator: Christiaan Huygens's classic observation of pendulum clocks synchronizing through a common wooden beam [Huygens, 1665], mode-locking in lasers, and Yoshiki Kuramoto's nonlinear synchronization model [Kuramoto, 1975] demonstrate how collective order emerges endogenously. In particle physics, this vision resonates with Geoffrey Chew's S-matrix "Bootstrap" program, where particle spectra form a self-consistent whole [Chew and Frautschi, 1961].

In this dynamical picture, a slowly changing gravitational field preserves the oscillon's resonantly locked mode structure, while its coordinate cadence Ω_t, operational size, and externally read mass adjust coherently. A rapid and strong change breaks the locking and redistributes energy into other compatible channels.

From One Oscillon to the Particle Spectrum. One localized oscillon family is the minimal self-contained dynamical problem of the particle sector. Its stable finite-energy form creates the spatial profile of a self-forming resonator; its small fluctuations reveal the internal s_i spectrum and connect it to the common environmental cadence Ω_t. The collective phase θ_C belongs to the organized collective state of the foundation, whereas the periodic oscillon phase belongs to its local cycle. Keeping these coordinates distinct places cosmological relaxation and particle rhythm within one coherent picture. The oscillon's energetic content and the response it leaves in the surrounding medium identify the physical variables of the bridge between rest mass and pole mass.

4.4 The Global Resonator: Unity of Micro- and Macro-Spectra

The two-level filter describes how the particle spectrum is selected; the global resonator supplies the arena in which that selection is realized.

Ernst Chladni's classic acoustic experiment provides the guiding intuition [Chladni, 1787]: sand scattered across a vibrating metal plate collects along the nodal lines of standing waves, where the oscillation amplitude is minimal. In the RefG map, this image corresponds to global modes and their nodes. Every resonator possesses its own spectrum: its geometry and boundary conditions determine which modes are amplified and which are suppressed.

After the operational network has formed, RefG reads it as a **unified global volumetric resonator**. Only at this stage does the concept of an eigenspectrum acquire a defined meaning: the connection operator, network geometry, and boundary conditions select the admissible modes. The central hypothesis is that the resonator's three-dimensionality and its spectrum are expressed simultaneously in physical particles and in the cosmic web. During subsequent expansion and thermal evolution, its modes may lock into stable physical forms:

- **Short-Wavelength Modes** at high frequencies correspond to localized elementary particles, or oscillons;
- **Long-Wavelength Modes** at low frequencies correspond to the large-scale distribution of matter, the **Cosmic Web**.

According to this hypothesis, the distribution of galaxies and galaxy clusters may resemble sand on a Chladni plate: filaments and vast cosmic voids may reflect the nodal pattern of standing modes in the global resonator.

Standard cosmology describes this large-scale morphology through Yakov Zel'dovich's anisotropic gravitational collapse [Zel'dovich, 1970]. RefG relates the same morphology to the ontological language of a single spectral operator.

The central hypothesis of this map is that **the microscopic elementary particle and the macroscopic cosmic web are two different readouts of one universal spectral operator**: one is a locally self-confined topological form, and the other is a globally extended cosmic pattern.

A joint empirical test of this operator spans:

1. The localized particle mass spectrum;
2. The cosmological history of structure formation;
3. The mass budgets of galaxy clusters;
4. The Baryon Acoustic Oscillation standard ruler at approximately $100\,h^{-1}$ Mpc [Eisenstein et al., 2005].

In the picture of a finite, boundaryless resonator, conceptually related to Christof Wetterich's variable-gravity framework [Wetterich, 2013], the geometry of the foundation expands according to its own internal rule, the background pressure falls, and material standards settle into their new state. A cosmological observer reads this causal chain through the ratio $A = a/p$. The expansion of the Universe is the evolution of its internal geometry, and the entire measurement procedure is defined within that geometry.

In current CMB observations, the low-multipole anomalies at $\ell = 2, 3$ remain of modest statistical significance. Searches for cosmic topology establish topology-dependent lower bounds and do not select a unique global form [Planck Collaboration, 2016].

4.5 Discreteness and the C_3 Generation Symmetry

In the Standard Model, the masses of the charged leptons—electron (e), muon (μ), and tau (τ)—are empirical Yukawa-coupling inputs. In the RefG map, they form the **ninth-order internal three-phase (C_3) triad** of a single topological defect: three distinct resonant modes of one underlying physical structure.

The charged-lepton three-phase parametrization is written most cleanly in square-root-mass space. Defining $x_k \equiv \sqrt{m_k}$ gives

$$x_k = \mu \left[1 + A_K \cos\left(\theta + \frac{2\pi k}{3} \right) \right], \qquad k = 0, 1, 2, \tag{41}$$

where:

- μ is the common scale in $\sqrt{m}$ space;
- A_K is the dimensionless three-phase modulation amplitude;
- θ is the phase angle of this algebraic mass map;
- $k = 0, 1, 2$ indexes the three generations (e, μ, τ).

When $A_K = \sqrt{2}$, the C_3 algebra identically recovers Yoshio Koide's empirical mass relation [Koide, 1982, 1983]:

$$K = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3}. \tag{42}$$

This three-phase parametrization possesses a rich literature:

- Carl Brannen formulated the relation in terms of circulant matrix eigenspaces [Brannen, 2006];

- Robert Foot demonstrated that $K = 2/3$ corresponds to a 45° geometric orientation in $\sqrt{m}$-space [Foot, 1994];
- Brannen's Pancharatnam–Berry phase construction yielded $\pi/12$ while leaving $\theta = 2/9$ as a dynamical parameter [Brannen, 2010];
- Konstantin Shulga obtained $\theta_\ell = -2/9$ within a compact family cycle [Shulga, 2026].

RefG connects this structure to a **ninth-order return cycle**. On the selected $h = 2$ branch, the first non-trivial topological return gives the mass angle $\theta = h/9 = 2/9$, while $A_K = \sqrt{2}$ expresses a geometric rule of equal energy sharing between the internal singlet and doublet vibrational channels.

A simple C_3-invariant potential is blind to θ. RefG locates the physical source of the phase angle in locked topological holonomy.

The tree-level point $(A_K, \theta) = (\sqrt{2}, 2/9)$ fails a strict numerical identification with the pole masses. The discrepancy points directly to the radiative pole-mass bridge between the geometric spectrum and observed masses [Sumino, 2009].

The acoustic intuition remains intact: the charged-lepton C_3/Koide triad can be read as a slightly anharmonic chord of a three-dimensional volumetric resonator. In this map, the x_k are coordinates of a candidate oscillon eigenspectrum; θ is the family mass angle, whereas θ_C is the collective phase of the foundation.

5 Topology and Quantum Numbers

In this chapter, RefG arranges the quantum numbers of the Standard Model within a unified topological research map. Phase, orientation, confinement, and holonomy are read as possible manifestations of a common geometric source for particle spin, charge, color, and generations. Covariant actions, symmetry algebras, and measurable amplitudes supply the dynamical language of these correspondences.

5.1 Phase, Orientation, and Topological Quantization

Thus far, the foundation-medium has been described primarily through vibrational amplitude and local pressure. Its complete local state is represented by several coupled field variables: the phase-clock scalar, amplitude profile, and spatial orientation frame each carry their own degrees of freedom. In this monograph, Φ is a compact designation for the complete configuration, simultaneously carrying amplitude, phase, internal geometric form, and spatial orientation.

Outside a defect, a single-valued scalar phase obeys $\mathbf{u} = \nabla\theta$ and $\nabla \times \mathbf{u} = 0$. The vortex-like phase-winding sector used here is characterized by:

- **a multivalued phase**,
- **a singular defect line**, and
- **an internal orientation locked in space**.

At the central core of the defect, the wave amplitude vanishes ($|\psi| = 0$), while the phase winds around it by an integer multiple of 2π ($\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n$). A canonical laboratory example is a quantized vortex filament in superfluid helium.

Its cosmological analogue is a **cosmic string**—a macroscopic topological defect line frozen in during an early-Universe phase transition, whose formation is described by the Kibble mechanism [Kibble, 1976]. In the language of RefG, it is a "frozen seam" in the foundation: vacuum topology determines whether such lines form, while astrophysical observations, including CMB lensing and pulsar timing, tightly constrain their cosmological density.

Lord Kelvin's vortex-atom program helped launch the modern mathematical theory of knots [Thomson, 1867, 1869, Tait, 1877]. Its historical outcome also established a clear boundary: topology supplies the kinematic skeleton of form, while dynamics selects a particle's physical viability. RefG joins these two sides by retaining the topological knot as the carrier of form and associating its stability with pressure–energy feedback and population-wide resonant locking among oscillons.

Phase quantization arises when the field is continuous along a loop, the state returns single-valuedly after a complete circuit, and the loop belongs to a nontrivial topological class ($\pi_1 \neq 0$).

For a closed, orientable, framed ribbon, the **Călugăreanu–White–Fuller (CWF) theorem** applies [Călugăreanu, 1961, White, 1969, Fuller, 1971, Ricca and Nipoti, 2011]:

$$Lk = Wr + Tw, \tag{43}$$

where

- Lk (linking number) is the topological integer measuring total linkage;
- Wr (writhe) describes the spatial coiling of the ribbon axis;
- Tw (twist) describes the internal phase twisting of the frame around that axis.

The geometric quantities Wr and Tw may vary continuously under a deformation that preserves the ribbon without self-intersection or cutting—for example, spatial coiling may be converted into internal twist—while their algebraic sum Lk remains strictly topologically invariant.

In hydrodynamics, helicity ($H = \int \mathbf{u} \cdot (\nabla \times \mathbf{u})\, d^3x$) is conserved in an ideal fluid, and Keith Moffatt connected it to the linking number Lk [Moffatt, 1969]. In superfluid helium, circulation quantization was predicted by Lars Onsager and Richard Feynman [Onsager, 1949, Feynman, 1955] and measured experimentally by Joe Vinen [Vinen, 1961]. Integer phase winding is therefore a directly measured laboratory phenomenon.

In the RefG map, the fundamental principle **"topological confinement = charge"** means that electric charge is the external readout of a confined phase–orientation path: the topological invariant specifies the discrete class, while field dynamics determines its electromagnetic readout.

In the synthetic map of RefG, **spin-1/2 and elementary charge** are read as two geometric readouts of one double-covered topological source: the half-twist determines the spinorial behavior of the state, while the chiral orientation of the twist and its integer winding determine the sign and magnitude of charge. Their geometric source may be common, while spin transformation and electric-flux normalization are fixed separately.

This direction has direct experimental analogues: Kelvin waves have been excited on quantized vortices in superfluid helium and their dispersion measured [Minowa et al., 2025]; helical axial modes and their dispersion have likewise been measured in a classical water vortex [Barckicke et al., 2026]. In RefG, the radial mode of an oscillon corresponds to its Compton-like mass pulsation, while the transverse helicity mode corresponds to the radiative and charge channel.

5.2 Spin and Fermionic Statistics

A phase field twisted by a half-turn (π) along an oscillon's internal axis generates a **Möbius-strip-type double-covered Z_2 structure**:

- one spatial circuit (2π) reverses the orientation and changes the sign of the wave amplitude ($\psi \to -\psi$);
- two complete circuits (4π) fully restore the initial state ($\psi \to +\psi$).

This geometry exactly reproduces the characteristic 4π return of spinorial behavior and supplies the topological skeleton of the 4π spinorial kinematics associated with spin-1/2. Its covariant physical realization combines an $SU(2)$ representation, a Lorentz spinor, and the corresponding action. Changing the topological class requires the amplitude to vanish in the core, the defect to reconnect, or the boundary condition to change.

The complete realization of fermionic statistics is defined by a phase of π and an antisymmetric Fock-space state under exchange of two identical oscillons:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1). \tag{44}$$

Under this condition, the total amplitude of two identical particles in the same quantum state vanishes ($\Psi = -\Psi \Rightarrow \Psi = 0$), yielding the Pauli exclusion principle. A bosonic mode retains its sign under exchange ($\Psi \to +\Psi$) and permits many bosons to occupy the same state.

In relativistic quantum field theory, the spin–statistics theorem unites Lorentz invariance, positive energy, and microcausality [Pauli, 1940]. The topological map of RefG supplies a geometric basis for this connection; its exact physical realization requires a covariant spinor sector satisfying the conditions of the spin–statistics theorem.

5.3 Charge, the Photon, and the Electromagnetic Channel

RefG reads **electric charge** as the external spatial trace of an oscillon's internal phase twist:

- right-handed orientation of the twist corresponds to positive charge ($+e$);
- left-handed orientation corresponds to negative charge ($-e$).

Externally, this trace extends radially in every direction. In the quantitative form of the electromagnetic channel, its total flux through a closed surface must obey Gauss's law:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}. \tag{45}$$

A **positron** is the complete topological mirror image of an electron in this map, with the opposite phase twist. During electron–positron annihilation ($e^+ + e^- \to 2\gamma$), the opposite windings cancel and open, transferring locally confined energy into the freely propagating transverse radiation channel. In a neutral oscillon, opposite phase twists balance one another and the net flux vanishes; this map corresponds to the neutrino sector.

A uniform rotation of the phase-asymmetry angle is a global $U(1)$ symmetry. Describing a spacetime-dependent rotation requires a local $U(1)$ phase connection and a covariant derivative. RefG reads the **photon** as a **transverse helicity mode** of this connection: it propagates a change in phase asymmetry through space and acts as the massless gauge carrier of the electromagnetic interaction.

Topological structure also appears in classical Maxwell electrodynamics: in Antonio Rañada's vacuum solutions, electric and magnetic field lines form mutually linked loops classified by the Hopf invariant [Rañada, 1989].

The sign of a force—attraction or repulsion—is determined jointly by the spin of the carrier, the type of source, and the coupling structure:

- **vector electromagnetic channel (spin-1):** like charges repel and opposite charges attract;
- **tensor gravitational channel (spin-2):** in a one-metric weak-field geometry, positive nonrelativistic sources couple with the same sign and attract at leading order; the full $T_{\mu\nu}$ determines the response to relativistic pressure, stress, and vacuum sources.

In three spatial dimensions, a conserved massless spherical flux dilutes as $1/r^2$, and its corresponding potential scales as $1/r$.

Hydrodynamic Bjerknes forces [Bjerknes, 1906] mark the precise boundary of this analogy: in an acoustic fluid, the phase relation of pulsations determines the sign of the force, whereas in electrostatics the sign follows from the structure of the vector gauge coupling. Carrier spin, source type, and topological coupling jointly determine attraction and repulsion.

RefG reads the fine-structure constant, $\alpha \approx 1/137$, as the dimensionless strength of this electromagnetic channel and as the invariant ratio between the electric-flux and transverse-photon responses of the foundation's nodal network.

5.4 Color Confinement and Three Generations

The three-dimensionality of space plays a fundamental role in the particle map of RefG: **the internal three-axis frame of an oscillon provides a natural geometric carrier for the color-charge triad—red, green, and blue (r, g, b).** Color is an internal degree of freedom of this frame.

- **Quark:** a topologically unclosed segment organized along one internal axis;
- **Baryon (proton, neutron):** a fully antisymmetric color-singlet binding of all three internal axes, forming a closed, stable, colorless configuration;
- **Meson:** a binding of an axis with its anti-axis—color with anticolor—likewise forming a closed, colorless state.

In this geometric map, **color neutrality** is represented by complete closure: a three-color baryonic binding and a color–anticolor mesonic pair form closed configurations, whereas an isolated one-axis color segment remains open. Color confinement is the dynamical realization of this structural rule.

The eight modes of a deformed three-axis oscillon numerically match the eight generators of $SU(3)$ and define a structural map of the color sector. The $SU(3)$ structure constants, local gauge symmetry, and interaction vertices provide the gauge-theoretic content of this map.

The **three-generation problem** is a logical extension of the C_3 symmetry discussed in the preceding chapter:

- charged leptons (e, μ, τ) represent the ninth-order internal three-phase modes of one topological defect;
- quark generations $(u/d, c/s, t/b)$ repeat the same generational architecture in the colored three-axis sector.

In this map, the three generations are read as successive internal holonomic return modes of the same volumetric resonator.

A standard quantum-mechanical realization of holonomy is **Michael Berry's geometric phase** [Berry, 1984]: during adiabatic cyclic evolution in parameter space, a quantum state accumulates a purely geometric holonomic phase alongside its dynamical phase. RefG associates this mechanism with the ninth-order return cycle of a topological defect; on the selected $h = 2$ branch, it gives $\theta = 2/9$ radians.

Why Is Space Three-Dimensional? In RefG, the three-dimensionality of space ($D = 3$) is directly connected with the **stability of topological knots**:

- closed one-dimensional loops form nontrivial, stable knots and links only in $D = 3$ spatial dimensions;
- in higher dimensions ($D \geq 4$), any knot can be opened and unlinked along an additional spatial coordinate without self-intersection.

This topological argument draws on the work of Paul Ehrenfest, Gerald Whitrow, and modern authors [Ehrenfest, 1918, Whitrow, 1955, Berera et al., 2017]. In the synthetic picture of RefG, the oscillon discreteness of matter and the three-dimensionality of space are read as two consequences of the same confinement condition.

5.5 Geometric Map of the Standard Model

RefG organizes the particle spectrum of the Standard Model into a single geometric map. The correspondences below establish a common structural language; the covariant action, symmetry algebra, and measurable amplitudes give these correspondences their dynamical content.

1. The Electron and Charged Leptons. In this map, the electron corresponds to a complete toroidal, Möbius-type twisted Z_2 structure in which all three spatial axes close locally. The double cover carries the 4π return characteristic of a Dirac spinor; its quantitative magnetic response belongs to the covariant spinor dynamics.

The measured electron g factor is slightly greater than two. Julian Schwinger calculated the first quantum correction, while modern Penning-trap experiments give $|g_e|/2 = 1.00115965218059(13)$ [Schwinger, 1948, Fan et al., 2023]. This number is among the most stringent quantitative targets for the topological map.

2. Quarks and Color Charge. In this map, quarks are axial topological segments with open ends. Their observed fractional electric charges $(+2/3, -1/3)$ are represented by one-third substeps of a three-dimensional closed loop, while baryons and mesons appear as colorless, fully closed configurations.

3. Gauge Bosons.

- **Photon:** a massless, propagating transverse $U(1)$ gauge mode with two helicity states;
- **Gluons:** eight massless transverse modes that mediate transitions among color states ($SU(3)$ octet);
- **$W^{\pm}$ and Z^0 bosons:** locally self-confined, massive carrier oscillon modes ($SU(2)_L \times U(1)_Y$).

RefG reads **beta decay ($n \to p + e^- + \bar{\nu}_e$)** as a topological rearrangement: a change in the orientation of an axial mode inside the neutron ($d \to u$) produces an intermediate charged weak mode (W^-), which decays into electron and neutrino channels. The chiral current, interaction vertex, decay amplitude, and measured spectrum determine the quantitative content of this picture.

4. Neutrinos and the Higgs Boson.

- **Neutrino:** an electrically neutral, incompletely confined chiral phase mode. Incomplete confinement is associated with its small nonzero, sub-eV mass scale and weak interactions, while Pontecorvo flavor oscillations correspond to phase mixing among internal mass modes;
- **Higgs boson:** a purely spherical radial breathing pulsation of the foundation-medium with no angular momentum, naturally corresponding to a spin-0 scalar state ($m_H \approx 125.25$ GeV).

5. The Graviton and the Weinberg–Witten Theorem. The TEGR-equivalent low-energy geometry yields the massless spin-2 sector of General Relativity and its two transverse-traceless polarizations, h_+ and $h_\times$. The Fierz–Pauli/Deser spin-2 bootstrap independently checks the same result. In the foundational map, these two polarizations correspond to modes of the nodal shear spectrum.

The Weinberg–Witten theorem [Weinberg and Witten, 1980] applies to a Lorentz-covariant quantum field theory defined on a Minkowski background with a local, gauge-invariant microscopic $T_{\mu\nu}$. The pregeometric foundation of RefG begins from a different structure, while the Lorentzian metric and Hilbert source emerge in the low-energy operational continuum. The graviton is read here as a collective tensor-shear excitation of the foundation-medium. If the completed microtheory satisfies every premise of the Weinberg–Witten theorem, this composite-graviton branch is excluded.

6. Nonlinear Field Lagrangian. The dynamical class corresponding to this topological map is a nonlinear field Lagrangian with stable, finite-energy knotted solutions; the Faddeev–Niemi model provides a well-known example of this class [Faddeev and Niemi, 1997].

5.6 Fundamental Constants: α and the Gravitational Hierarchy

The fine-structure constant:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.035999} \tag{46}$$

remains one of the greatest mysteries in physics. Richard Feynman regarded it as one of the deepest unexplained numbers in theoretical physics [Feynman, 1985].

In RefG, α is the **dimensionless invariant ratio between the electric-flux and transverse-photon responses of the foundation-medium's nodal network**.

Three strict filters guide the search for the origin of α:

1. Simple ratios of Planck and electron masses or frequencies do not yield 137;
2. The identity $r_e/\bar{\lambda}_c = \alpha$, relating the classical electron radius to the reduced Compton wavelength, is exact but definitional;
3. Integer phase-locking steps are real physical phenomena, as the Josephson effect and Shapiro steps demonstrate [Josephson, 1962, Shapiro, 1963], but they do not by themselves determine the numerical value of α.

RefG relates α to the ratio of the intrinsic response strengths of the electric topological flux and the photon's transverse oscillation. A common scale transformation readjusts both responses coherently, leaving their dimensionless ratio invariant.

The Hierarchy Problem: Why is Gravity So Weak? The electrostatic repulsion between two protons exceeds their gravitational attraction by a factor of 10^{36}, while for electrons this ratio reaches 10^{42}:

$$\frac{F_{\text{em}}}{F_g} = \frac{e^2}{4\pi\epsilon_0 G m_e^2} = \alpha\left(\frac{M_{\text{Pl}}}{m_e}\right)^2 \sim 4.17\times 10^{42}. \tag{47}$$

This immense disparity motivated Paul Dirac's Large Numbers Hypothesis [Dirac, 1937].

RefG associates this fundamental hierarchy with the different physical structures of the two interaction channels:

1. **The electromagnetic channel** carries a direct phase-transverse gauge coupling between the topological charges of oscillons;
2. **The gravitational channel** is the long-range refractive response to the Bernoulli-type static pressure deficit generated in the surrounding medium by an oscillon's internal oscillatory energy ($-\Delta P \propto \rho_{\text{dyn}}$).

- RefG reads the Planck mass ($M_{\text{Pl}} = \sqrt{\hbar c/G} \approx 2.18\times 10^{-8}$ kg), whose rest-energy scale is $M_{\text{Pl}}c^2 \approx 1.22\times 10^{19}$ GeV, as a natural calibration scale for the elastic response of the foundation;
- The electron and proton are localized particle modes with energies many orders of magnitude below the Planck scale ($m_e/M_{\text{Pl}} \sim 10^{-22}$).

The algebraic identity $\alpha(M_{\text{Pl}}/m)^2$ rewrites the observed hierarchy in a single expression. Its value is diagnostic: if the foundation action produces G, the Planck calibration scale, and the oscillon's particle energy within one dynamics, then the small ratio m/M_{Pl} suppresses the gravitational response quadratically.

Thus RefG interprets the hierarchy through the distinct structures of the two channels: **electromagnetic coupling is the direct phase response of charge, whereas gravity is pressure-mediated refraction generated by microscopic oscillons.**

6 Epilogue: Boundaries of Knowledge

6.1 Why One Foundation and One Manifested Medium?

At the conclusion of this work, we return to the central philosophical and physical question: **Why one foundation-medium?**

Ontological parsimony is the first advantage: a common carrier reduces the number of independent initial axioms and assumptions. The deeper argument is the **common origin of the regularities of nature**.

When the electron, quark, and photon are read as different resonant and topological modes of one universal foundation-medium, physical interactions become couplings among modes of a common carrier, while conservation of energy, momentum, spin, and charge becomes the geometric expression of its symmetries and shared energy balance.

This argument has an important historical backdrop. In his celebrated essay, Eugene Wigner wrote of the "unreasonable effectiveness of mathematics" in the natural sciences [Wigner, 1960]. The picture of one ontological carrier seeks a physical explanation for that mysterious order: **the shared mathematical regularities of the Universe reflect the internal dynamical structure of a single foundational substance.**

The central question is therefore: *What produces the common symmetry rule that binds different fields into one set of universal laws?* In the picture of one foundation and one manifested medium, this order comes from the internal dynamics of the shared carrier. What appear as coincident laws are read as different expressions of a deeper structure.

The argument also has a stringent empirical face—the problem of **exact structural correspondences**:

- the exceptionally tight observational bound on the electrical neutrality of macroscopic matter ($|q_p + q_e|/e < 10^{-21}$);
- the observational compatibility of the photon rest mass with zero ($m_\gamma < 10^{-18}$ eV);
- the rigid system of fractional quark charges [Navas et al., 2024].

These facts call for a common structural explanation, which RefG associates with the topological geometry of the modes of a single foundation.

One universal foundation can produce a rich and varied phenomenology: it can carry volumetric pressure, shear stress, phase, topology, vortical rotation, memory, and resonance at once. This picture echoes Philip Anderson's celebrated formula—**"More is Different"**: each new level of organization has physical regularities of its own [Anderson, 1972]. RefG reads the passage between these levels as the self-organization of a multichannel foundation.

RefG reads the measurable cosmological Universe as a **self-measuring global resonant network**. The foundation itself is pre-spatial and is assigned no spatial size. A finite, connected periodic network with no unique center provides a mathematical example of this possibility. The actual dimension, topology, finiteness, and boundarylessness of the Universe are fixed by global spectral and observational signatures. The essential physical claim remains: the Universe measures itself through its own resonant modes.

6.2 Three Operational Boundaries

The Universe is always measured by components of the Universe itself: our clocks, measuring rods, detectors, and even the observer's body are oscillon structures of the universal foundation-medium. This concept of a **self-measuring Universe** resonates with John Wheeler's *It from Bit* program [Wheeler, 1990]. Wheeler gives primacy to information; RefG gives primacy to the physical foundation-medium and reads the informational trace as a result of its self-distinction.

Here a *boundary* denotes the domain of validity of a measurement procedure. An observer within the Universe can measure only those relations defined in the active nodal network itself.

Kurt Gödel's incompleteness theorem [Gödel, 1931] serves here as a fundamental epistemological analogy: just as a consistent formal system contains true propositions that cannot be proved from within the system, a self-measuring Universe raises the question: *How completely can an internal part of a system measure the global boundaries of the whole?*

This self-measuring picture gives rise to three operational boundaries:

1. **The Operational Origin of Time:** In the selected scenario, time begins with the first foundation-wide self-distinction, assigned the boundary values $t = 0$ and $\tau = 0$. No physical past exists before Genesis, because neither change nor an ordering of events has yet been defined. Background parameters determine the post-Genesis history, while a unified bridge among the photon, atomic standards, and operational geometry makes this history accessible to an internal observer. On the selected integrable p branch, the total process age is finite; its numerical value is fixed by the background parameters, Genesis matching, and clock calibration.

2. **The Operational Boundary of Space:** We measure the connections among distinguishable oscillon nodes. An exterior of the Universe is not a possible location of observation. The cosmological horizon is a measurement boundary defined from within the network. The quantity $R_{\rm act}$ is a local activation or correlation scale and does not denote an external radius of the Universe.

3. **The Operational Boundary of Scale:** In RefG, the Planck length, $\ell_{\rm Pl} \sim 10^{-35}$ m, is the natural transition scale built from c, $\hbar$, and G, at which quantum and gravitational effects become important together. The minimum operational distinguishability threshold ℓ_* is a separate physical concept; its relation to the Planck length belongs to the microdynamics of the foundation.

In all three cases, the boundary arises from the conditions of measurement internal to the Universe. A finite object may acquire an unbounded coordinate image, as in the stereographic projection of a sphere onto an infinite plane.

6.3 Methodological Principles

The methodological principle of this work is simple: **an intuitive physical picture must be transformed into a rigorous, computable, self-consistent, and falsifiable formalism.** Along this path, each particle hypothesis passes through five successive gates:

1. **Empirical Target:** mass, charge, spin, lifetime, form factor, and the relevant uncertainty are fixed in advance;

2. **Geometry and Dynamics:** the resonant-topological class is selected, and its spatial profile, boundary conditions, and nonlinear action are written explicitly;

3. **Symmetry and Stability:** Noether currents, quantum numbers, zero modes, the finite-energy solution, and the perturbation spectrum are tested within one system;

4. **Parameter Closure and Prediction:** free coefficients are fixed independently of the target data, after which a new measurable result is calculated;

5. **Decisive Test:** the prediction is compared with data under a predefined falsification criterion.

In this sequence, intuition sets the direction of research, while calculation establishes its physical force. Every quantitative result is tied to a specific equation, the data used, and a reproducible computation.

The minimal complete unit of a particle derivation is one finite-energy oscillon. Its action determines the localized profile; the perturbation operator determines its eigenspectrum; the two-time dictionary determines coordinate and process frequencies; and an independent energy–mass map connects that spectrum to measurable rest and pole masses.

6.4 Where the Theory's Path Ends: A Falsification Map

A defining virtue of a scientific theory is its falsifiability (Popper's criterion). The falsification map of RefG is organized into three strict hierarchical levels:

Falsification of the Entire Framework
Lorentz violation, equivalence-principle violation, transposition failure

$\Downarrow$

Falsification of an Individual Sector
wave budget, MOND/RAR, EHT

$\Downarrow$

Falsification of a Specific Geometric Reading
Koide, h/m, $g = 2$

1. The Entire Framework Is Falsified If:

1. The oscillon sector fails to recover effective Lorentz invariance and spatial isotropy of light speed ($|\Delta c/c| > 10^{-17}$; Herrmann et al. 2009), or gravitational-wave propagation speed deviates from c ($|v_{\mathrm{gw}} - c|/c > 10^{-15}$; Abbott et al. 2017).
2. At the same fixed total energy, inertial and gravitational mass exhibit a composition-dependent discrepancy exceeding observational bounds ($\eta > 10^{-15}$; MICROSCOPE: Touboul et al. 2022).
3. The global pressure evolution of the foundation-medium fails to yield a unique relation between process and metric time, or the cosmology built from that relation fails to recover the observed CMB, BBN, photon redshift, and Type Ia supernova $(1+z)$ time-dilation relations [Blondin et al., 2008].
4. A dimensionless ratio assigned to the common adiabatic-transposition class—for example α—exhibits a measurable change under a local pressure shift ($|\Delta\alpha/\alpha| > 10^{-6}$; Kotuš et al. 2017).
5. Self-distinction of the foundation fails to establish a unified bridge to mutually distinguishable modes, local causality, and additive accounting of energy, momentum, and charge.

2. An Individual Sector Is Falsified If:

1. Gravitational-wave energy flux fails to reproduce binary pulsar orbital decay to a precision of 10^{-3}–10^{-4} [Kramer et al., 2021], or scalar breathing modes exceed LIGO-Virgo-KAGRA bounds.
2. The galactic vortex framework fails to reproduce SPARC/RAR data [McGaugh et al., 2016] or the observed spatial separation and lensing amplitude of the lensing centroids relative to the X-ray gas in the Bullet Cluster [Clowe et al., 2006].
3. A complete Einstein–Boltzmann calculation sourced by the neutral collective phase fails to recover the primary CMB anisotropies, or its late-time continuation fails to reproduce the BAO $100\,h^{-1}$ Mpc ruler and large-scale-structure statistics [Eisenstein et al., 2005].
4. The compact regular-core model fails to match Event Horizon Telescope shadow diameters, the compact-branch oscillation spectrum, or gravitational-wave echo limits [Cardoso and Pani, 2019].

5. The neutral phase-density realization fails to admit an independent cyclic coordinate θ_C, the common phase shift cannot be maintained as an exact symmetry, or the foundation and operational densities of one charge Q_C fail the exact normalized Jacobian $\widehat{n}_{C,\text{op}} = p^3 \widehat{n}_{C,F}$.

3. A Specific Geometric Reading Is Falsified If:

1. The electron oscillon profile fails to maintain a pointlike electromagnetic form factor down to 10^{-18} m [Navas et al., 2024].
2. The Koide/C_3 branch fails to construct a radiative bridge to pole masses, or a fourth stable charged lepton is discovered.
3. Galactic-flow circulation is not quantized in units of h/m within an independently fixed effective-mass and coherence window.
4. The Möbius double-cover fails to yield the Dirac value $g = 2$, Fock-space antisymmetry, or the corresponding quantum corrections.
5. The calculated fluctuation spectrum of one oscillon fails to exhibit common Ω_t factorization; in that case, universal particle transposition and the mass ratios inferred from it are removed from the program.

Each criterion has a concrete observational or computational measure: a negative answer from nature removes the corresponding idea from the research program rather than merely changing its verbal formulation.

Conclusion

The central idea of Refractive Gravity can be stated in one sentence: **geometry, matter, clocks, measuring rods, and light are different operational readouts of one state of the foundation.** The Universe is therefore no longer described as a stage detached from its observer; the observer is part of the same physical process being measured.

The Genesis picture preserves this unity from the beginning. The first foundation-wide self-distinction of the pre-spatial foundation produces event order, relation, and time. The resonant fields of localized oscillatory seeds become coupled, while stable closures form matter. An oscillon generates a static pressure deficit in its surroundings; that deficit jointly changes the cadence of time, the spatial standard, and the motion of bodies. Time dilation, scale transformation, and gravitational geometry are three readings of this single response.

This intuition has a precise mathematical bridge. Under specified post-Genesis continuum conditions, the leading action of the relational coframe takes TEGR form, and the torsion–curvature identity relates it to the Einstein–Hilbert action up to a boundary term. When matter couples universally and minimally to this one metric and no additional unsuppressed mode enters at 1PN order, the complete standard 1PN response follows. RefG extends Einstein's geometry with a physical ontology of the foundation and reads spacetime curvature as an operational manifestation of the foundation's state.

The same causal chain takes a different form as the scale changes. In compact objects, a deep pressure deficit is read as strong-field geometry; on galactic scales, coherent vortical flow provides an additional dynamical channel; on cosmological scales, expansion of the foundation lowers its background pressure. An internal observer cannot measure this expansion directly as a change in inter-node separation, because the observer's clock and measuring rod are made from the same foundation. The process is read through the change of the material standard, while the single operational geometry $A = a/p$ unifies redshift, signal time dilation, and distances into one history. Expansion and contraction of the standard are not two additive effects: the latter is the internal measurement trace of the former.

On the same A geometry, the cold branch of the neutral collective phase current has exactly the background and linear behavior of a pressureless Einstein source. Under the standard local atomic/QED branch, gravity-only coupling to the plasma, and an adiabatic initial mode, the CMB initial-value problem takes the standard Einstein–Boltzmann, recombination, and line-of-sight form. RefG extends the physical interpretation of that established calculation by giving the geometry and the cold source a common foundation origin.

At the particle level, the volumetric resonator, common cadence, and topological invariants form a unified geometric map of mass, spin, charge, color, and generations. The first dynamical bridge of this map is now in place: the complex phase action admits a finite-energy, charge-protected Q-ball-type core, while its strict-vacuum quadratic action takes exactly the free complex-scalar form and yields the corresponding quantum field under standard canonical quantization. Thus the coframe that produces Einstein–Hilbert geometry also carries standard free quantum matter on the same metric.

Thus RefG presents the Universe as a self-measuring resonant system. The state of the foundation produces operational geometry; this geometry guides the motion of matter and light; and clocks and measuring rods made from that same foundation read the entire process from within. In this single circuit, gravitation, cosmological evolution, the structure of matter, and the origin of measurability become parts of one physical history.

Current Status of the Research Program

This section gathers the exact state of the program in one place. Each entry first states the result obtained, then its domain of validity and the formal or observational gate that turns the corresponding sector into a complete theory.

1. **Ontological Foundation, Cosmic Genesis, and the Origin of Measurability**—The current Genesis scenario unites a centerless initial boundary common to the entire foundation, the first minimal relations, localization of oscillon seeds, overlap of resonant fields, coupling-mediated network connection, and the subsequent formation of stable oscillons. Formal checks separate the parameter domains, the causal ordering of events, and the conditional balance of energy-transfer signs after the initial event. A finite periodic graph combines connectivity and the absence of a unique center in one construction; by contrast, strict radial ordering from a single site in a symmetric network creates a preferred center and falls outside the centerless scenario. The complete origin mechanism comprises the transition action, the actual dimension and topology of the foundation, an oscillon-seed solution, the resonant-field equation, the physical coupling threshold, the initial energy distribution, the thermal history, and the observational consequences that follow from them.

2. **Inter-Node Separation and Minimum Distinguishability**—The principle of operational distinguishability has been formulated: physical distance acquires meaning only after a distinguishable relation has formed. A Planck-order threshold is the probable physical scale of this transition. Its exact value and relation to the Planck length are fixed by the microdynamical action of the foundation.

3. **Local Common Cadence**—The relations $\Omega_t/\Omega_{t,0} = p$ and $d\tau = p\,dt$ provide the exact dictionary between coordinate and process time. The particle decomposition $\nu_i^{(t)} = s_i\Omega_t$ is an independent spectral hypothesis: the environment fixes the common cadence Ω_t, while the internal structure of the oscillon fixes the dimensionless labels s_i. Its decisive formal gate is the perturbation operator of one finite-energy oscillon, whose eigenspectrum must factor out the common Ω_t.

4. **Oscillon Deficit, Size, Time, and Mass**—The metric filters and the first-order biconformal branch have been obtained symbolically. On this branch, operational size, process cadence, and externally read mass follow the same p response. This sector now contains a minimal coframe-coupled complex $U(1)$ action, its analytic existence window, and a finite-energy radially nodeless Q-ball-type candidate core on the fixed non-backreacting Minkowski coframe, supported

by convergent numerical evidence of orbital stability. Full dynamical closure now requires coframe backreaction and an exact map connecting the core's internal energy to operational size, pressure deficit, process cadence, rest mass, and pole mass.

5. **Relational Geometry, the Einstein–Hilbert Action, and 1PN**—In a static, asymptotically uniform, nonrotating, spherically symmetric weak field, the common clock–ruler response gives $ds^2 = p^2c^2dt^2 - p^{-2}d\boldsymbol{\sigma}^2$, $c_{\text{local}} = c_0$, and $\gamma = 1$. A canonical gradient term, linear coupling to the active source density entering Gauss's law, and the absence of an additional unsuppressed term at the same order determine the additive logarithmic response $u = -\ln p$, from which $u = GM/(c^2r)$ and $\beta = 1$ follow.

   The post-Genesis continuum description is built on a $3+1$ relational coframe, one operational metric, flat inertial transport, a local and reversible leading action in the constant-coefficient parity-even quadratic-torsion class, minimal phase coupling, the absence of unsuppressed mixed operators at the same order, and the inertial-gauge character of orientation. In the regular linear Minkowski limit of this class, the TEGR coefficient ratios follow. The exact torsion–curvature identity then converts the action to Einstein–Hilbert form up to a boundary term.

   On the positive, stable, and causal phase branch, $\rho_C > 0$, $\rho_C' > 0$, and $0 \le n_{C,\text{op}}\rho_C''/\rho_C' \le 1$. Metric variation gives $T^C_{\mu\nu}$, while variation of θ_C gives a conserved current whose homogeneous operational reduction yields $n_{C,\text{op}}A^3 = \text{const}$. The Jacobian $n_{C,\text{op}} = p^3 n_{C,F}$ reads the same law in foundation measure as $\eta_F a^3 = 1$. In the complete action, every physical sector enters the total $T^{\text{tot}}_{\mu\nu}$ exactly once through its own metric variation. The Fierz–Pauli/Deser spin-2 self-coupling route independently checks the same Einstein–Hilbert operator.

   The complete standard PPN vector follows through 1PN order when the full action consists of the Einstein–Hilbert operator and matter universally and minimally coupled to the same metric, the source is represented by one complete conserved stress–energy tensor, RefG and GR use the same local source, PPN gauge, background, and boundary conditions, and no additional unsuppressed field or operator enters at this order: $\gamma = \beta = 1$ and $\xi = \alpha_1 = \alpha_2 = \alpha_3 = \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0$. Its domain comprises isolated, weak, slowly moving, and weakly self-gravitating sources.

   The complete bridge to the foundation comprises the origin of the nodal graph and its dimension, the construction of a smooth coframe and its Lorentzian time direction, the emergence of flat transport and the common cone, the gauge character of orientation, the coframe–phase split, the microscopic oscillon $T_{\mu\nu}$, the microphysics of $\rho_C(n_{C,\text{op}})$ and its relation to $n_{C,F}$ and P_F, the numerical values of G and Λ_F, the 2PN and strong-field regimes, and particle-source matching. The same bridge must also determine the error incurred in passing from the discrete foundation to the continuum limit and the microscopic mechanism that suppresses mixed, nonminimal, and higher-derivative terms.

6. **Inertia and Gravitational Mass**—For a finite-energy Lorentz-covariant localized solution satisfying the Laue boundary condition $\int T^{ij}\, d^3x = 0$ for all i, j, the Noether identity relates the inertial coefficient and the gravitational source to the same total energy of the system. The next formal gate is a direct derivation of this relation from the oscillon's translational zero mode.

7. **Gravitational Waves, Local Unobservability, and the Domain of the Composite Graviton**—The TEGR-equivalent one-metric geometry gives the two standard transverse-traceless (TT) polarizations propagating on the common light cone and the corresponding detector response. The nodal spectrum of the foundation must supply the microscopic origin of the common cone, the two TT modes, and the gauge character of orientation; at the same level it fixes cosmological propagation, source–flux normalization, the complete waveform, and the fate of possible additional polarizations.

   The Weinberg–Witten theorem applies to a Lorentz-covariant quantum field theory defined on a Minkowski background with a local, gauge-invariant microscopic $T_{\mu\nu}$. The composite-graviton map of RefG is based on a pregeometric foundation and a Lorentzian metric that emerges at low energy. If the completed microtheory satisfies all Weinberg–Witten premises simultaneously, this composite-graviton branch is excluded.

8. **Compact Objects and the NEC Gate**—A static map and finite-core matching of a trial C^2 profile have been obtained. Under one specific exterior-layer and pressure-reversal criterion, the admissible domains do not overlap. This exposes a clear dynamical tension: the stable phase current $\rho_C + p_C = n_{C,\mathrm{op}}\rho_C' > 0$ belongs to a class of sources compatible with the null energy condition (NEC), whereas the regular compact-core map requires an effective NEC-violating contribution. A complete source for the compact sector must rest on an equation of state that reconciles these requirements and then determines collapse, rotation, and full stability.

9. **Galaxy Rotation, Morphology, and Clusters**—The selected Ginzburg–Landau-type effective potential
$$W(g_v, g_N) = \frac{g_v^3}{3} + \frac{g_N g_v^2}{2} - a_0 g_N g_v$$
gives $g_N = g^2/(g + a_0)$ at equilibrium, and in the deep limit gives $g^2 \simeq a_0 g_N$ together with the baryonic Tully–Fisher relation $v^4 = GM_b a_0$. In a filtered SPARC sample of 2,788 points from 163 galaxies, with $\Delta V/V < 10\%$, $V_{\mathrm{obs}} > 20\ \mathrm{km\,s^{-1}}$, fixed a_0, $\Upsilon_{\mathrm{disk}} = 0.5$, $\Upsilon_{\mathrm{bulge}} = 0.7$, and no galaxy-by-galaxy parameter fitting, the residual root-mean-square scatter is 0.141 dex and the mean residual is -0.012 dex. This establishes compatibility between the selected effective map and the data.

   In a QUMOND estimate on a 300×300 grid, with $a = 3$ kpc and $b = 300$ pc, the maximum correction to the radial acceleration relative to the algebraic map is 9.23% over $a \le r \le 5a$. The ideal logarithmic branch gives the lensing angle $\alpha = 2\pi v_f^2/c^2$; on this branch, rotation and lensing are two readings of the same effective potential.

   Doppler cancellation, $c_{\mathrm{coord}}/c_0 = p^2$, follows from symbolic algebra. Independent predictive status requires the origin of W and a_0 from the foundation, the process dynamics of vortical rotation, and a predefined test against PHANGS/SPARC data.

   The morphological map associates disk and spiral structure with ordered flow, and elliptical structure with a more isotropic or chaotic regime. The cluster mass budget combines three channels, each counted once: a long-wavelength nodal deficit, the remote pressure fields of galaxies, and cluster-scale vortical memory. The decisive measures of this map are circulation quantization $\Gamma = nh/m$, dynamical recovery of the observed spatial separation and lensing amplitude in the Bullet Cluster, and the complete cluster mass budget. Primary CMB anisotropy, the BAO $100\,h^{-1}$ Mpc ruler, and large-scale-structure statistics test the unity of this galactic map with the cosmological phase source.

10. **Cosmological Relaxation, the Phase-Current Source, the Time Bridge, and the Arrow of Time**—The ratio $A = a/p$ is the internal geometric readout of one expansion–pressure–standard history. In the post-Genesis homogeneous foundation description, $\Psi_C = \sqrt{n_{C,F}}e^{i\theta_C}$ represents a neutral collective phase state. Conservation of the Noether phase charge in a comoving three-dimensional foundation volume gives $\eta_F a^3 = 1$ exactly, while the same charge is read in the operational geometry as $n_{C,\mathrm{op}}A^3 = \mathrm{const}$.

   In the action for a barotropic, isentropic, irrotational phase current, variation of θ_C gives $\nabla_\mu J^\mu = 0$, the barotropic energy function defines $p_C = n_{C,\mathrm{op}}\rho_C'(n_{C,\mathrm{op}}) - \rho_C(n_{C,\mathrm{op}})$, and metric variation gives the Hilbert stress–energy tensor $T^C_{\mu\nu}$. Here p_C is the thermodynamic pressure of the phase sector, whereas P_F is the operational readout of the background pressure of the foundation; their exact relation must be fixed by a constitutive equation derived from the microphysics of the foundation. On the selected homogeneous, isotropic branch comoving with the network, one obtains $P_F/P_{F,0} = a^{-3}$, $p = a^{-3/2}$, and $A = a^{5/2}$.

On the same branch, the inverse map gives the exact relations $a = A^{2/5}$, $p = A^{-3/5}$, and $\eta_F = P_F/P_{F,0} = A^{-6/5}$. This is a unique reconstruction of the normalized relative scale. For an ideal-comoving pair, the fixed step count N_{12} of the selected path records relational structure, whereas Q_C records the conserved charge of the phase sector; the two records play independent physical roles.

The same A determines redshift, signal time dilation by $(1+z)$, the Hubble function, and geometric distances. At equal background parameters, these quantities form one operational equivalence class with the corresponding ideal observables of spatially flat ΛCDM; distinguishing physics must appear in source and clock microphysics, calibration, or inhomogeneous dynamics. In this background comparison, the two descriptions are statistically equivalent. A constant vacuum contribution enters the total $\Lambda_{\rm eff}$ term exactly once.

The linear formal CMB bridge is closed on the selected one-metric, spatially flat, linear post-Genesis continuum and the cold phase branch with fixed energy per conserved charge, under the conditions that standard matter, the Maxwell field, neutrinos, and the local atomic/QED sector couple minimally to the common metric; local dimensionless atomic and QED ratios retain their standard values; $T^C_{\mu\nu}$ couples to the plasma only gravitationally; and the initial transfer mode is the standard unit-normalized adiabatic mode. The normalized densities of one conserved charge Q_C obey $\widehat{n}_{C,\rm op} = p^3\widehat{n}_{C,F} = A^{-3}$, where $\widehat{n} \equiv n/n_0$ and $A(t_0) = a(t_0) = p(t_0) = 1$. This exact identity belongs to the homogeneous background, while perturbations evolve directly through the operational current. The relation $\rho_C = \mu_C n_{C,\rm op}$ gives the exact cold source $p_C = c^2_{s,C} = \sigma_C = 0$. Under these conditions, the once-only source registry, background equations, linear scalar dynamics, Thomson collisions, neutrino hierarchy, recombination, optical depth, and line-of-sight endpoint take the standard Einstein–Boltzmann form. The `cdm` computational slot may represent $T^C_{\mu\nu}$; no separate particle cold source is added. At this stage, Ω_{C0} is an input normalization for the CMB forward calculation.

The resulting status is an **exact conditional handoff from** RefG **to the standard Einstein–Boltzmann, recombination, and line-of-sight CMB initial-value problem**. This is closure at the level of equations, not a claim that the spectra have already been computed. The subsequent observational stage comprises connecting both the amplitude and scale dependence of the primordial $P_{\mathcal{R}}(k)$ to Genesis; fixing possible isocurvature and tensor initial conditions; deriving the microphysical normalization of Ω_{C0}; specifying reionization; numerically executing a Boltzmann code; calculating primary and lensed $TT/TE/EE$ spectra; and performing the likelihood and full data comparison. That stage determines the observational CMB status; the theoretical initial-value problem is already in standard computational form.

The time bridge $d\tau = p\,dt$ relates coordinate and process time. Both begin with the first distinguishable relation at the boundary $t = 0$, $\tau = 0$; an integrable p gives a finite process history. RefG associates the global relaxation of the foundation with the physical direction of the macroscopic arrow of time. Quantitative closure requires an entropy law and irreversible foundation dynamics that connect the thermodynamic, cosmological, and radiative arrows of time to one source.

The complete microphysical bridge for the cosmological sector comprises the nodal-oscillon origin of $\rho_C(n_{C,\rm op})$, its relation to $n_{C,F}$ and P_F, the normalization of Ω_{C0} and its possible common origin with the galactic a_0, the foundation origin of local atomic/QED ratios, the microscopic coframe–phase split, particle sources, concrete inhomogeneous solutions, the numerical value of Λ, Genesis matching, and a complete data model.

11. **Two-Level Time and Quantum Randomness**—The interpretive map provides a picture compatible with Bell correlations and delayed-choice facts: the internal coordination of the foundation belongs to one unified state, while propagating physical signals obey local causality. Three results determine the closure of this sector as a formal theory: derivation of the Born

distribution $P = |\psi|^2$, exact preservation of the no-signaling condition, and a complete dynamics of measurement.

12. **Particles as Oscillon Modes and the Scalar Quantum-Field Bridge**—The particle sector already contains the local nonlinear amplitude–phase action and a finite-energy spherical Q-ball-type candidate core with numerical evidence of orbital stability. The strict-vacuum quadratic sector of that same action takes exactly the form of the free complex Klein–Gordon action. Under the explicitly declared standard canonical quantization rule, its fixed Minkowski branch gives the free excitations about the strict vacuum a free Hamiltonian bounded below and nonnegative after vacuum normalization, bosonic Fock space, particle and antiparticle sectors, a global $U(1)$ phase charge, microcausality, a propagator pole on the mass shell, the spin-zero Casimir, and a controlled Schrödinger limit. This is the first exact RefG bridge to standard quantum field theory within the same coframe that yields Einstein–Hilbert geometry.

    The next particle gate is collective quantization of the nonlinear localized core and its identification as a one-particle, bound, or coherent state. It is followed by the dynamics of population interlocking, the full spectrum of stability and decay channels, proton-decay lifetime bounds, tests of common frequency factorization and the t/τ dictionary, and the transition from rest mass to the interacting pole mass. Deriving $\hbar$, the canonical commutators, Hilbert space, the Born rule, and the vacuum choice from the foundation is a separate quantum-foundation bridge. At this status, the global phase charge remains an internal $U(1)$ charge, and the free spin-zero quantum remains a free excitation about the strict vacuum.

13. **The Koide, C_3, and 2/9 Structure**—In $x_k = \sqrt{m_k}$ space, the C_3 algebra gives $K = 2/3$ exactly for $A_K = \sqrt{2}$. With $(A_K, \theta) = (\sqrt{2}, 2/9)$ and the electron pole mass as anchor, it misses the muon mass by 451.7σ and the tau mass by 0.61σ. Thus C_3 is a sharply defined external algebraic structure. Establishing its physical realization requires oscillon dynamics, the corresponding eigenspectrum, and an action-derived radiative pole-mass bridge.

14. **Spin, Pauli Exclusion, and Fermionic Statistics**—The Möbius Z_2 double cover exactly reproduces the kinematics of a sign change under 2π and return to the initial state under 4π; it supplies the topological skeleton of the 4π spinorial kinematics associated with spin-1/2. The formal gate for the complete spinor sector is the joint recovery of an $SU(2)$ representation, a Lorentz spinor, and a covariant action that yields the Dirac equation and the factor $g = 2$.

    The bridge to fermionic statistics requires an exact phase of π under exchange of two identical oscillons, Fock-space antisymmetry, and the Pauli principle. A complete relativistic realization joins these results to Lorentz invariance, positivity of energy, microcausality, and the corresponding quantum corrections.

15. **Charge, the Photon, and the Electromagnetic Channel**—The following structural results have been obtained: a local $U(1)$ phase symmetry, two helicities, screening of the asymptotic radial σ-field component—which would make the energy grow linearly with radius if left unscreened—and, after screening, a surviving $1/r^2$ far field, flux accounting, and subtheorems for a conserved phase current. Complete formal closure of the sector comprises canonical charge normalization, Maxwell's equations, Gauss's law, Ward identities, quantum electrodynamics (QED), the Dirac sector, the exact Hilbert source of freely propagating radiation, and the pressure-geometric response it generates.

16. **Color, Quarks, and Gluons**—The three-color, eight-mode construction provides a partially compatible structural map of the color sector. The numerical match between eight modes and the eight generators of $SU(3)$ establishes a structural correspondence. Full gauge identity requires recovery of the $SU(3)$ structure constants, local $SU(3)$ gauge invariance, interaction vertices, scale-dependent evolution, the color-confinement potential, and canonical normalization of fractional charge.

17. **The Electroweak Map, Beta Decay, and the Higgs**—The current result is a geometric placement map of the electroweak sector of the Standard Model. Its dynamical closure comprises $SU(2)_L$ chirality, a W/Z mass matrix that preserves a massless photon, CKM/PMNS mixing, the neutrino mass ordering, Yukawa couplings, and the Higgs vacuum expectation value. The topological-rearrangement picture of beta decay becomes a quantitative theory when the action yields the chiral current, interaction vertex, transition amplitude, decay rate, energy–momentum balance, and the complete electron–neutrino spectrum.

18. **The Fine-Structure Constant** α—A nodal interpretation and a target-independent spectral measure have been obtained: α is read from the relative strengths of the electric-flux and photonic-response channels, while the common scale cancels. A complete derivation of the physical value must unite a compact microscopic action for the physical charge Q, a unique dimensionless ratio, the full charged spectrum, and evolution to the Thomson limit. The earlier boundary formula remains a target-dependent conditional result.

19. **The Gravitational Hierarchy**—The force ratio is written algebraically as $F_{\rm em}/F_g = \alpha(M_{\rm Pl}/m)^2$ and displays the correct scale hierarchy. RefG associates the difference with the contrast between direct phase-transverse electromagnetic coupling and a secondary pressure-refractive gravitational response. A fundamental derivation must unite G, $M_{\rm Pl}$, the microscopic energy of an oscillon, and the particle mass hierarchy within one dynamics.

20. **The Self-Measuring Universe and the Micro–Macro Spectral Bridge**—The global resonator is a central cosmological-level hypothesis. A finite periodic graph places connectivity and the absence of a unique center within one mathematical construction. In this picture, the short-wavelength eigenmodes of one universal spectral operator are read as particle oscillons, while its long-wavelength modes are read as the structure of the cosmic web and galaxy clusters.

    The decisive test of micro–macro identity must connect, at once, the localized particle mass spectrum, the history of cosmological structure formation, the mass budget of galaxy clusters, and the BAO standard ruler. The finiteness, boundarylessness, dimension, and topology of the Universe require evidence beyond the periodic-graph example; low-multipole and topological signatures remain observational candidates for this global geometry.

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